RELATIVE HODGE SPECTRA OF MULTISCALE CONIFOLD SMOOTHINGS

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Abstract. We determine the complete small two-form spectrum of the Ricci-flat metrics on an explicit compact polarized Calabi–Yau conifold smoothing with eight nodes. The estimates hold on its full six-dimensional period base, with no restriction on the relative sizes of the nonzero local smoothing parameters. Exactly two positive eigenvalues tend to zero. A fixed exterior cohomological map identifies their spectral space with the two-dimensional quotient of divisor classes by those surviving the smoothing, and compares its entire energy form relatively with the capacity-weighted restriction form. The harmonic mass and complementary spectral gaps are controlled as well. The proof constructs a uniform Ricci-flat correction through independently scaled Stenzel regions, retaining their quadratic harmonic modes by exact complex automorphisms, and combines weighted capacity bounds with compactness in degrees one and two. The resulting energy and mass forms give uniform asymptotics for both eigenlines in these cohomological coordinates and yield separated logarithmic and iterated-logarithmic spectral scales on the same algebraic family.

1. Introduction

The small spectrum of a degenerating metric records both the topology of the degeneration and the energy required to join its limiting harmonic fields. When several local scales tend to zero independently, an estimate at the largest small eigenvalue can lose the smaller modes. We prove a relative comparison for the complete small two-form spectrum of actual Ricci-flat metrics on a compact Calabi–Yau conifold family. The family has eight nodes, two lost divisor directions, and six independent vanishing-period coordinates. It permits arbitrary separation of the two positive spectral scales.

The central threefold and its small resolution are fixed throughout. They arise from the anticanonical contraction of

$$Z = \mathbb{P}_{\mathbb{P}^1\times\mathbb{P}^1}(\mathcal{O}\oplus\mathcal{O}(1,-1)) \times_{\mathbb{P}^1\times\mathbb{P}^1} \mathbb{P}_{\mathbb{P}^1\times\mathbb{P}^1}(\mathcal{O}\oplus\mathcal{O}(1,-1)). \tag{1}$$

For a general anticanonical hypersurface $Y \subset Z$, chosen also nondegenerate with respect to its toric polytope, the induced morphism $q : Y \to X_0$ contracts eight disjoint $(-1,-1)$ curves. They form two groups I_f, I_g of four curves of the same respective homology types. This compact construction and its Picard data are established in [W26b, Appendix A]. We retain the exact polarization $\mathcal{L}_0 = (-K_{\bar{Z}})|_{X_0}$, where $\bar{Z}$ is the anticanonical model. Section 2 describes the contraction and proves the required multiparameter smoothing statement.

Date: November 12, 2026
This version: November 12, 2026.
2020 *Mathematics Subject Classification.* 58J50 (primary); 14J32, 32Q25, 53C21 (secondary).
Key words and phrases. Calabi–Yau threefold, conifold transition, Hodge Laplacian, small eigenvalues, Ricci-flat metric, relative spectral comparison.

There is a ball B about zero in

$$R = \left\{ p \in \mathbb{C}^8 : \sum_{i \in I_f} p_i = \sum_{i \in I_g} p_i = 0 \right\} \tag{2}$$

and a polarized compact family $\pi : \mathcal{X} \to B$ with central fibre X_0. A nowhere-zero relative holomorphic three-form Ω has oriented local vanishing periods p_i. In node coordinates the equations are

$$z_1^2 + z_2^2 + z_3^2 + z_4^2 = \tau_i(p), \qquad p_i = \tau_i(p) u_i(p), \tag{3}$$

where the u_i are holomorphic and bounded above and away from zero. The smooth locus is exactly $B^\circ = \{p \in B : \prod_i p_i \neq 0\}$. Thus the node parameters are controlled by their periods even when one is smaller than every prescribed power of another. A first-order description of the deformation space would not give this conclusion.

Let g_p be the Ricci-flat Kähler metric in $c_1(\mathcal{L}_p)$ on X_p, for $p \in B^\circ$. We use the real Hodge Laplacian $\Delta_p = \mathrm{d}\delta_p + \delta_p \mathrm{d}$, with its usual closed-manifold operator domain. Define

$$\kappa_i(p) = \frac{2\pi}{|\log|\tau_i(p)||}, \qquad w_\alpha(p) = \sum_{i \in I_\alpha} \kappa_i(p), \qquad E_p = \begin{pmatrix} w_f & 0 \\ 0 & w_g \end{pmatrix}, \qquad M = \frac{4}{5} \begin{pmatrix} 9 & -1 \\ -1 & 9 \end{pmatrix}. \tag{4}$$

The energy is the capacity-weighted squared restriction of the lost divisor classes to the links. The mass is the central harmonic mass after orthogonal projection off all surviving cohomology. Both are constructed independently of the actual positive small eigenvalues.

To specify the comparison of eigenspaces, fix a compact central exterior C_* by deleting node balls of a small positive radius. Its central metric defines the L^2 projection Π_* onto distributionally closed two-forms, with unrestricted tangential trace. Let $V = H^2(Y;\mathbb{R})$, let G be its central harmonic mass, and let $K \subset V$ be the two-dimensional subspace of classes surviving on the smooth fibres. Put $W = K^{\perp_G}$. The cohomology of the exterior identifies its closed projection with a class in V. The observation map is

$$\mathcal{A}u = \mathrm{pr}_W^G[\Pi_*(u|_{C_*})]. \tag{5}$$

Here and below the restrictions to C_* use fixed smooth exterior identifications. The closed projection and its cohomology map are bounded on exterior L^2 forms. Formula (5) therefore has a precise meaning for form-domain inputs; it does not take arbitrary cycle integrals of an unclosed L^2 form. We use the G-orthogonal lifts of the quotient classes f, g as coordinates on W.

Theorem 1.1. *After shrinking B, there are $\gamma > 0$ and a function $e(p) \to 0$ as $p \to 0$ in B° with the following properties.*

The harmonic two-form space of (X_p, g_p) has dimension two. Its cohomological identification with K transports its mass to a form converging to $G|_K$. Exactly two positive two-form eigenvalues lie below γ. If $\mu_1(p) \leq \mu_2(p)$ are the generalized eigenvalues of (E_p, M), these eigenvalues satisfy

$$(1 - e(p))\mu_j(p) \leq \lambda_{2,j}^+(p) \leq (1 + e(p))\mu_j(p), \qquad j = 1, 2. \tag{6}$$

The first positive one-form eigenvalue and the third positive two-form eigenvalue are at least γ.

The two-dimensional positive small spectral space S_p consists of coexact primitive real $(1,1)$ forms. The restriction $\mathcal{A} : S_p \to W$ is an isomorphism. Its transported mass and energy,

$$N_p(x,y) = \langle \mathcal{A}^{-1}x, \mathcal{A}^{-1}y\rangle_{g_p}, \qquad Q_p(x,y) = \langle \mathrm{d}\mathcal{A}^{-1}x, \mathrm{d}\mathcal{A}^{-1}y\rangle_{g_p}, \tag{7}$$

satisfy the quadratic-form inequalities

$$(1-e(p))M \le N_p \le (1+e(p))M, \qquad (1-e(p))E_p \le Q_p \le (1+e(p))E_p. \tag{8}$$

All conclusions hold throughout B° as $p \to 0$, with no bound on the ratios of the nonzero node parameters or of the positive model eigenvalues.

The whole-form conclusion retains mixed pairings in the prescribed coordinates. If u_1, u_2 are actual orthonormal eigenforms, C_p has columns $\mathcal{A}u_j$, and $\Lambda_p = \mathrm{diag}(\lambda^+_{2,1}, \lambda^+_{2,2})$, it gives

$$\|C_p^T M C_p - I\| \le e(p), \qquad \|\Lambda_p^{-1/2}(C_p^T E_p C_p - \Lambda_p)\Lambda_p^{-1/2}\| \le e(p), \tag{9}$$

after changing the error function by a fixed factor. In particular, an off-diagonal energy error is bounded by a vanishing multiple of $\sqrt{\lambda^+_{2,1}\lambda^+_{2,2}}$. No explicit rate is claimed for the compactness-derived part of $e(p)$.

The finite eigenvalues are

$$\mu_\pm = \frac{9(w_f+w_g) \pm \sqrt{81(w_f-w_g)^2 + 4w_f w_g}}{128}. \tag{10}$$

They satisfy $0 < \mu_-/\mu_+ \le 4/5$. The corresponding actual eigenvalues are consequently simple for all sufficiently small parameters. A relative subspace estimate identifies each actual eigenline, through $\mathcal{A}$, with its finite generalized eigenline, with error tending to zero uniformly in w_f/w_g.

For example, the admissible path

$$p(h) = (a,a,a,-3a;b,b,b,-3b), \qquad a = e^{-h^{-1}}, \quad b = e^{-h^{-2}}, \quad h \downarrow 0, \tag{11}$$

has actual asymptotics

$$\lambda^+_{2,1}(p(h)) \sim \frac{10\pi}{9}h^2, \qquad \lambda^+_{2,2}(p(h)) \sim \frac{9\pi}{8}h. \tag{12}$$

More generally $a = e^{-F(h)}$, $b = e^{-G(h)}$, with positive $F, G \to \infty$ and $F/G \to 0$, gives coefficients $9\pi/(8F)$ and $10\pi/(9G)$. These are paths in a holomorphic multiparameter family; they need not extend as holomorphic arcs at zero.

1.1. **Relative reduction and earlier results.** Relative small-spectrum comparison has substantial precedents. Burger's graph theorem for degenerating hyperbolic surfaces already permits unequal short-geodesic lengths and gives multiplicative scalar eigenvalue asymptotics [Bur90, Theorem 1.1]. In cusp surgery, Albin–Rochon–Sher identify positive small Hodge modes by a restriction difference and compute their leading rescaled energy [ARS18, Section 3, Lemma 3.2 and Theorem 3.3]. Anné–Takahashi's conic analysis retains the cap and critical extended solutions needed for a complete spectral count [AT15, Theorems A–B]. The geometry and parameter domains of these theorems differ from the polarized Ricci-flat smoothing used here.

For actual Ricci-flat conifolds, Hein–Sun establish the central conical metric asymptotics [HS17, Theorem 1.4]. Benabida proves polyhomogeneous smoothing expansions along prescribed rays [Ben26, Theorem C]. Pacini's scalar gluing estimates already allow independent scales on distinct cap components [Pac13, Theorems 11.7 and 12.3]. We verify the scalar decomposition, initial Kähler family, global inverse and nonlinear correction on the whole

period domain, then prove the complete two-form comparison there. The fixed-ray spectrum in [W26a] supplies the central geometric normalization, not the unequal-scale theorem by substitution.

The metric proof first identifies all fifteen scalar harmonic modes of growth below three on a Stenzel cap. Nine are pluriharmonic; the remaining six are derivatives of complex orthogonal automorphisms. A global scalar inverse retains these modes separately from a positive-weight remainder. In the nonlinear equation we use the finite automorphisms themselves: their exact preservation of the Stenzel volume removes the nondecaying quadratic terms. This gives the actual Ricci-flat comparison on every cap and intervening annulus without a smallest-scale loss.

For the two-form problem, corrected logarithmic trials have exactly the capacity energy, including zero contribution from a link on which the period vanishes. A weighted estimate on each independent cap prevents low-energy mass concentration in degrees one and two. A fixed exterior closed projection then gives a lower bound in the input's own energy norm. Multiplicative metric comparison transfers the full count and gaps to the actual metrics, while the exterior projection preserves the cohomological marking.

The abstract argument in Section 3 isolates three geometric estimates: linear trials, a weighted lower energy inequality for the specified observation, and capture of low-energy mass. It allows the exact kernel to change and obtains relative energy on the entire observed low space. The final step uses a determinant on the quotient by the common kernel. It is useful precisely when an additive mass-plus-energy comparison would lose the weakest direction. The geometric estimates, rather than the finite matrix factorization alone, are the analytic content of an application.

The hierarchy of the finite forms retains the capacities, the full restriction map and the harmonic mass. Its relation to the classical attachment sequence is explicit. It is distinct from the Leray filtration of a fixed adiabatic fibration [MM90], and from the corrected foliation spectral-sequence statements of Álvarez López–Kordyukov [AK00]. No new nearby-cycle functor or canonical integral lattice on a general small eigenspace is asserted.

2. The compact family and its period coordinates

2.1. **The central transition.** Write $B_0 = \mathbb{P}^1_f \times \mathbb{P}^1_g$, $L = \mathcal{O}_{B_0}(1,-1)$, and use the quotient convention in (1). Let ξ_1, ξ_2 be the two tautological divisor classes and put $k_i = g + \xi_i$. Each k_i is globally generated, since the direct image of its line bundle is $\mathcal{O}(0,1) \oplus \mathcal{O}(1,0)$. If $[s_0 : s_1], [t_0 : t_1]$ are coordinates on B_0, the resulting map to $A = \mathbb{P}^3 \times \mathbb{P}^3$ has local formula

$$(x; y) = (v_1 s; u_1 t), \qquad (x'; y') = (v_2 s; u_2 t), \tag{13}$$

where (u_i, v_i) are the quotient coordinates. This formula respects the changes of trivialization of L, and pulls back $\mathcal{O}_A(1,0)$ and $\mathcal{O}_A(0,1)$ to k_1 and k_2.

Its image is the complete intersection

$$V_0 = \{x_0 x_1' - x_1 x_0' = 0, \quad y_0 y_1' - y_1 y_0' = 0\} \subset A. \tag{14}$$

Indeed the two determinant equations specify the common directions s, t whenever the corresponding pairs are nonzero. One direction is free exactly when both its pairs vanish; both cannot be free at the same point of A. The remaining scalars then reconstruct (13). This proves set-theoretic surjectivity and irreducibility of the image. The two equations form a regular sequence and are generically smooth, so their complete intersection is reduced and has the asserted scheme structure.

The singular locus consists of the two disjoint lines

$$\ell_x = \{x = x' = 0,\ [y] = [y']\}, \qquad \ell_y = \{y = y' = 0,\ [x] = [x']\}. \tag{15}$$

Near ℓ_x, use nonzero y, y' coordinates and eliminate the second equation. The first is a 2×2 determinant in the four x coordinates, with a free coordinate along the line. The same argument applies to ℓ_y. Thus the transverse singularities are ordinary threefold nodes. The complete intersection is Cohen–Macaulay and regular in codimension one, hence normal.

The exceptional loci of $Z \to V_0$ are the two sections $S_f, S_g \simeq B_0$ on which both quotients are respectively onto $\mathcal{O}$ and onto L. Each contracts one ruling to its corresponding line. Their codimension in Z is two, so the map is small. Adjunction and the projective-bundle formula give

$$\omega_{V_0} = \mathcal{O}_{V_0}(-2,-2), \qquad -K_Z = 4g + 2\xi_1 + 2\xi_2 = 2k_1 + 2k_2. \tag{16}$$

The contraction is therefore crepant. Normality gives $q_{Z*}\mathcal{O}_Z = \mathcal{O}_{V_0}$, and projection formula identifies its anticanonical section ring with that of the ample line bundle $\mathcal{O}_{V_0}(2,2)$. In particular $V_0 = \bar{Z}$ is the anticanonical model in the central construction.

Choose a general anticanonical $Y \subset Z$, also nondegenerate for the toric polytope, and let X_0 be its image. The defining section of $\mathcal{O}_{V_0}(2,2)$ lifts to A: the twisted two-equation Koszul resolution is

$$0 \longrightarrow \mathcal{O}_A \longrightarrow \mathcal{O}_A(1,1)^{\oplus 2} \longrightarrow I_{V_0}(2,2) \longrightarrow 0,$$

and its higher cohomology vanishes. Hence X_0 is also described by the two equations in (14) and one equation of bidegree $(2,2)$. Its section meets each line in (15) in four simple points.

The same multiplicities and the exceptional periods can be computed on Z. The normal bundles and anticanonical restrictions of the two sections are

$$N_{S_f/Z} = (L^{-1})^{\oplus 2}, \quad N_{S_g/Z} = L^{\oplus 2}, \qquad (-K_Z)|_{S_f} = \mathcal{O}(0,4), \quad (-K_Z)|_{S_g} = \mathcal{O}(4,0). \tag{17}$$

The all-$\mathcal{O}$ and all-L summands of

$$\pi_*\mathcal{O}_Z(-K_Z) = \mathcal{O}(0,4) \otimes \mathrm{Sym}^2(\mathcal{O} \oplus L) \otimes \mathrm{Sym}^2(\mathcal{O} \oplus L)$$

have independent coefficients. The restrictions are consequently the complete quartic systems. At a simple zero the normal derivative of the equation removes the trivial summand in the normal bundle of the contracted ruling. Each resulting curve has normal bundle $\mathcal{O}(-1) \oplus \mathcal{O}(-1)$ in Y.

The global toric calculation in [W26b, Appendix A] gives $h^{1,1}(Y) = 4$, $h^{2,1}(Y) = 60$ and the spanning divisor classes (f, g, ξ_1, ξ_2). Simple connectivity follows from the toric lattice-quotient calculation in [W26a, Section 8]. We use these established global data. In these coordinates the two curve types have degrees

$$\Gamma_f = (1,0,0,0), \qquad \Gamma_g = (0,1,-1,-1). \tag{18}$$

The contraction described above has no other exceptional locus. The polarization retained in this paper is

$$\mathcal{L}_0 = \mathcal{O}_{V_0}(2,2)|_{X_0}, \qquad H = c_1(q^*\mathcal{L}_0) = (4g + 2\xi_1 + 2\xi_2)|_Y. \tag{19}$$

In particular it is not replaced by a multiple when an embedding is chosen below.

2.2. **Local periods through a node.**

Lemma 2.1. *Let $Q(z) = z_1^2+\cdots+z_4^2$ and let $Q(z) = \tau(s)$ be a holomorphic family of ordinary double points. For a nowhere-zero relative dualizing form Ω, the oriented local vanishing period is a single-valued holomorphic function $P(s) = \tau(s)u(s)$, with u holomorphic and $u(0) \neq 0$.*

Proof. Write

$$\Omega = a(z,s)\operatorname{Res}\frac{\mathrm{d}z_1\wedge\cdots\wedge\mathrm{d}z_4}{Q(z)-\tau(s)}, \qquad a(0,0)\neq 0,$$

choosing a holomorphic ambient representative for a. For positive real τ, the real vanishing sphere has radius $\sqrt{\tau}$. Scaling $z = \sqrt{\tau}\,y$ multiplies the residue form by τ. With the radial residue orientation its integral is $\pi^2\tau$: the residue is $(2\sqrt{\tau})^{-1}$ times the induced sphere volume, and the sphere volume is $2\pi^2\tau^{3/2}$.

Expand a in its convergent Taylor series in z. Odd homogeneous terms integrate to zero under the antipodal map of S^3, which preserves orientation. A term of degree $2k$ contributes τ^{k+1} times a holomorphic function of the remaining parameters. Uniform convergence on a smaller coordinate and parameter domain therefore gives $P = \tau u$, with $u(0) = \pm\pi^2 a(0,0)$. The two choices of square root give the same integral by the same antipodal symmetry. Analytic continuation is consequently single-valued, and the formula extends holomorphically across $\tau = 0$. □

2.3. **The six-dimensional polarized smoothing.**

Proposition 2.2. *There is a proper flat family $\pi : \mathcal{X} \to B$ over a ball $B \subset R$ as in (2), projective over B, whose central fibre is X_0. It has a relatively ample line bundle restricting to the exact $\mathcal{L}_0$ and a nowhere-zero relative holomorphic three-form. Its total space is smooth. The local equations, oriented periods and bounded units satisfy (3). Its complete discriminant is $\bigcup_i\{p_i = 0\}$; every smooth fibre is simply connected and has $b_2 = 2$.*

Proof. The central variety is Gorenstein Calabi–Yau, with rational local complete-intersection singularities. Rationality of the nodes and the Calabi–Yau resolution give $H^1(\mathcal{O}_{X_0}) = H^2(\mathcal{O}_{X_0}) = 0$. Its singularities are Du Bois and the projective resolution satisfies the $\partial\bar\partial$ lemma. Friedman's unobstructedness theorem [Fri26, Theorem 1.6] therefore gives a smooth abstract deformation germ.

Choose an integer N such that $\mathcal{L}_0^N$ is very ample and $H^1(\mathcal{L}_0^N) = 0$. Over a small Artin extension, $H^2(\mathcal{O}_{X_0}) = 0$ lifts the embedding line bundle and $H^1(\mathcal{L}_0^N) = 0$ lifts its sections. The lifted morphism remains a closed immersion over the nilpotent thickening. Thus embedded deformations map formally smoothly to abstract deformations. The restricted Euler sequence expresses the same fact as $H^1(T_{\mathbb{P}^M}|_{X_0}) = 0$. A neighbourhood S of the central point in the Hilbert germ is smooth and its tangent map onto the abstract deformation tangent is surjective. Shrink it to retain integral fibres.

The use of $\mathcal{L}_0^N$ for this embedding does not change the prescribed polarization. After an étale neighbourhood, the universal family has a section through a smooth central point [Sta, Tag 055S]. The relative Picard scheme and its Poincaré bundle normalized along this section are available by [Kle05, Theorems 2.5(2), 4.8; Exercises 3.11, 4.3 and their solutions]. Since $H^2(\mathcal{O}_{X_0}) = 0$, the Picard morphism is smooth at $[\mathcal{L}_0]$ [Kle05, Proposition 5.19]. A further étale neighbourhood has a section through that point. Pulling back the normalized Poincaré bundle extends $\mathcal{L}_0$ itself, and it is relatively ample after shrinking. Both base changes preserve the tangent image.

The parametric holomorphic Morse lemma gives node functions $\tau_i : S \to \mathbb{C}$. The local-to-global tangent calculation of Rollenske–Thomas [RT09, Section 3, (3.2)] identifies the image of their joint differential, up to nonzero choices of local generators, with the kernel of the exceptional curve-class map. By (18) this kernel has dimension six and contains a vector with every coordinate nonzero. Each $\mathrm{d}\tau_i(0)$ is therefore nonzero. The total space is smooth at the nodes and elsewhere near the central fibre. Properness permits a shrink making it smooth throughout.

The flat family is Gorenstein. Its relative dualizing line bundle restricts to $\omega_{X_0} \simeq \mathcal{O}_{X_0}$. Since $H^1(\omega_{X_0}) = 0$, cohomology and base change lifts a nonzero central generator to a relative section. Its zero set has proper image missing the origin. A further shrink gives a nowhere-zero relative form Ω.

We next determine the exact, rather than first-order, period relations. Delete disjoint node balls to obtain a common exterior U. Fixed collars and the proper submersion theorem with boundary identify its nearby exteriors, including parameters with some singular caps. The Thom and Mayer–Vietoris sequences give

$$H^2(Y;\mathbb{Z}) \xrightarrow{r} \mathbb{Z}^8 \xrightarrow{\partial} H^3(X_s;\mathbb{Z}), \qquad r(D)_i = D \cdot C_i, \quad \partial e_i = \mathrm{PD}_L(\delta_i). \tag{20}$$

Here PD_L uses the oriented Thom class of the vanishing sphere. To identify the maps, the exceptional curves have real codimension four, so restriction gives $H^2(Y) \simeq H^2(U)$. A resolution cap retracts to S^2, and restriction to its link records the curve degree. A smoothing cap retracts to S^3 and has no degree-two cohomology. Its boundary generator maps to the Thom dual of that zero section. These observations give exactness and the displayed maps.

The restriction vectors of f, g are the two block indicators v_f, v_g. Integral Poincaré duality therefore gives

$$\sum_{i \in I_f} \delta_i = 0, \qquad \sum_{i \in I_g} \delta_i = 0 \quad \text{in } H_3(X_s;\mathbb{Z}). \tag{21}$$

The four divisor classes span rational $H^2(Y)$, so these span all rational relations. No assertion about saturation of the vanishing image is needed.

Orient the local spheres compatibly with (20). Lemma 2.1 gives holomorphic periods $P_i = \tau_i u_i$. On the smooth locus, (21) gives $\sum_{I_f} P_i = \sum_{I_g} P_i = 0$. The smooth locus is dense: the smooth base integrates a full-support smoothing tangent, or, equivalently, the finite union of the nontrivial hypersurfaces $\{\tau_i = 0\}$ has empty interior. Holomorphic extension gives these identities on all of S.

Consequently $P = (P_i)$ maps S to R, and

$$\mathrm{d}P(0) = \mathrm{diag}(u_i(0))\mathrm{d}\tau(0)$$

has rank six. The holomorphic submersion theorem gives a section $\sigma : B \to S$, with $\sigma(0) = 0$ and $P\sigma = \mathrm{id}$. Pull back the compact family, polarization and relative form along σ. Lemma 2.1 now gives $p_i = \tau_i(p)u_i(p)$. After shrinking the ball, all units are bounded above and away from zero.

Properness excludes new singularities outside the chosen node charts. Inside chart i, the fibre is singular exactly when $\tau_i = 0$, equivalently $p_i = 0$. On that hyperplane $\mathrm{d}\tau_i = u_i^{-1}\mathrm{d}p_i \neq 0$, since the coordinate functional p_i is nonzero on R. The pulled-back total space is therefore smooth as well. The discriminant need not be a normal-crossings divisor at intersections of its hyperplanes.

Finally the links $S^2 \times S^3$ and both kinds of fillings are simply connected. Van Kampen identifies the fundamental group of a smooth fibre with that of Y, which is trivial. The

exceptional curve map has rank two, so the surgery sequence gives $b_2(X_p) = b_2(Y) - 2 = 2$. This proves all assertions. □

2.4. **Intersection data and permitted scales.** The Chow relations and normalization are

$$f^2 = g^2 = 0, \qquad \xi_i^2 = (f-g)\xi_i, \qquad \int_Z fg\xi_1\xi_2 = 1.$$

Integrating $D_iD_j(-K_Z)^2$ gives

$$B_{ij} := \int_Y D_iD_jH = \begin{pmatrix} 0 & 8 & 4 & 4 \\ 8 & 0 & 12 & 12 \\ 4 & 12 & -8 & 0 \\ 4 & 12 & 0 & -8 \end{pmatrix}, \qquad (D_i) = (f, g, \xi_1, \xi_2), \qquad H^3 = 320. \tag{22}$$

The kernel of restriction is spanned by $k_1 = g + \xi_1$, $k_2 = g + \xi_2$, and $H = 2k_1 + 2k_2$. Relative to these surviving classes and the quotient lifts f, g, the intersection blocks are

$$B_{KK} = \begin{pmatrix} 16 & 24 \\ 24 & 16 \end{pmatrix}, \quad B_{DK} = \begin{pmatrix} 12 & 12 \\ 12 & 12 \end{pmatrix}, \quad B_{DD} = \begin{pmatrix} 0 & 8 \\ 8 & 0 \end{pmatrix}. \tag{23}$$

The primitive Hodge pairing on the orthogonal quotient is consequently

$$-B_{DD} + B_{DK}B_{KK}^{-1}B_{KD} = \frac{4}{5}\begin{pmatrix} 9 & -1 \\ -1 & 9 \end{pmatrix}. \tag{24}$$

The analytic identification of this pairing with the limiting quotient mass uses harmonic representatives and removal of all surviving classes. The intersection calculation itself introduces no chosen actual small eigenform.

The two-parameter slice $(a, a, a, -3a; b, b, b, -3b)$ lies in R. Every pair of sufficiently small nonzero a, b gives a smooth fibre, with no comparison condition on their magnitudes. Boundedness of the units gives

$$\big|\log|\tau_i(p)| - \log|p_i|\big| \le C. \tag{25}$$

For $a = e^{-F(h)}$, $b = e^{-G(h)}$, where $F, G \to \infty$, this yields $w_f \sim 8\pi/F$ and $w_g \sim 8\pi/G$. It also explains why finite vanishing orders in one holomorphic arc behave differently: if $\tau_i(s) = c_i s^{\nu_i} + O(s^{\nu_i+1})$, then

$$\kappa_i(s) \sim \frac{2\pi}{\nu_i|\log|s||}.$$

All such capacities have one logarithmic order. The exact divisibility $\tau_i = p_i/u_i$ is what permits the more separated paths without uncontrolled higher-order additive terms in their smallest coordinates.

3. Relative reduction through a specified observation

We first give the spectral argument used to pass from geometric estimates to the complete low space. Its hypotheses concern trial functions, observed energies and low-energy mass. They do not assume the desired eigenvalue asymptotics or an identification of the low eigenspace.

3.1. **Forms and geometric estimates.** Let p range over a parameter space with a specified limit. Auxiliary parameters may vary without converging, provided the following estimates are uniform. For each p, let $\mathcal{H}_p$ be a real Hilbert space and q_p a densely defined nonnegative closed quadratic form with compactly embedded form domain. Let L_p be its associated self-adjoint operator, and order its eigenvalues, including zeros, as $\lambda_0(p) \le \lambda_1(p) \le \cdots$.

Independently of this spectrum, suppose that geometry supplies a real space V_p of dimension $1 \le n_p \le N$, a positive definite mass M_p, and a nonnegative energy E_p. We use the same letters for their polarizations. Write

$$\sigma_p = \sup_{x \neq 0} \frac{E_p(x)}{M_p(x)} \longrightarrow 0, \qquad \mu_0(p) \le \cdots \le \mu_{n_p-1}(p) = \sigma_p \tag{26}$$

for the largest and all the generalized eigenvalues of (E_p, M_p). There are three assumptions.

Linear trials. There is a linear map $T_p : V_p \to \operatorname{Dom} q_p$ such that

$$\|T_p x\|^2 \ge (1 - a_p) M_p(x), \qquad q_p(T_p x) \le (1 + a_p) E_p(x), \qquad a_p \to 0. \tag{27}$$

Thus T_p is injective for sufficiently small parameters. A trial construction depending on a fixed auxiliary cutoff is enough if, for each error tolerance, one proves a uniform parameter threshold and one linear trial map on that domain. A diagonal choice of cutoffs gives (27); smooth dependence of that choice is not needed.

Weighted observation. A specified linear map $\mathcal{A}_p : \operatorname{Dom} q_p \to V_p$ satisfies

$$E_p(\mathcal{A}_p u) \le (1 + b_p) q_p(u), \qquad b_p \to 0. \tag{28}$$

The error multiplies the entire energy. An additive $o(\sigma_p)$ bound is insufficient.

Low-energy mass capture. If S is spanned by at most $n_p + 1$ orthonormal eigenvectors with eigenvalue at most s, then

$$\left| M_p(\mathcal{A}_p u) - \|u\|^2 \right| \le \omega(p, s) \|u\|^2 \quad (u \in S), \qquad \lim_{s \downarrow 0} \limsup_{p \to 0} \omega(p, s) = 0. \tag{29}$$

We take ω nonnegative and nondecreasing in s. A bound for every form-domain vector of Rayleigh quotient at most s is stronger than required. In the geometric applications, this estimate includes the analysis that excludes additional concentration modes.

No positive lower bound is imposed on the nonzero ratios μ_j/σ_p. The kernels of E_p and L_p may change dimension. The three assumptions must hold uniformly through the asserted changes.

Theorem 3.1. *Under (26)–(29), there are a fixed $\gamma > 0$ and $e_p \to 0$ such that the spectral subspace S_p of L_p below γ has dimension n_p, and*

$$(1 - e_p)\mu_j(p) \le \lambda_j(p) \le (1 + e_p)\mu_j(p) \quad (0 \le j < n_p), \qquad \lambda_{n_p}(p) \ge \gamma. \tag{30}$$

The map $\mathcal{A}_p : S_p \to V_p$ is an isomorphism and maps $\ker L_p$ onto $\ker E_p$. Define

$$N_p(x, y) = \langle \mathcal{A}_p^{-1} x, \mathcal{A}_p^{-1} y \rangle_{\mathcal{H}_p}, \qquad Q_p(x, y) = q_p(\mathcal{A}_p^{-1} x, \mathcal{A}_p^{-1} y). \tag{31}$$

Then, as forms on the entire V_p,

$$(1 - e_p) M_p \le N_p \le (1 + e_p) M_p, \qquad (1 - e_p) E_p \le Q_p \le (1 + e_p) E_p. \tag{32}$$

The error is controlled by the dimension bound and the three hypothesis errors. It contains no condition number of the positive part of E_p.

Proof. Put $c_p = (1+a_p)/(1-a_p)$. Applying min–max to the linear trial map gives

$$\lambda_j \le c_p \mu_j \quad (j < n_p), \qquad \lambda_{n_p-1} \le c_p \sigma_p. \tag{33}$$

If there were no uniform complementary gap, a sequence tending to the limit would have $\lambda_{n_p} \to 0$. Choose $s > 0$ small enough that $\limsup_{p\to 0} \omega(p,s) < 1/2$. Eventually the first $n_p + 1$ eigenvectors have eigenvalue below s and $\omega(p,s) < 3/4$. Equation (29) would make $\mathcal{A}_p$ injective on their span, contradicting $\dim V_p = n_p$. This proves the gap. Together with (33), it identifies precisely n_p low eigenvalues.

Let $d_p = \omega(p, c_p\sigma_p)$. For each fixed $s > 0$, eventually $c_p\sigma_p \le s$, and hence $\limsup_{p\to 0} d_p \le \limsup_{p\to 0} \omega(p,s)$. Letting $s \downarrow 0$ shows $d_p \to 0$, also when $\sigma_p = 0$. The mass-capture estimate makes $\mathcal{A}_p$ injective on S_p, hence an isomorphism, and gives

$$(1+d_p)^{-1} M_p \le N_p \le (1-d_p)^{-1} M_p. \tag{34}$$

The trial map sends $\ker E_p$ injectively into $\ker L_p$. Conversely, the observation inequality sends $\ker L_p$ to $\ker E_p$, and mass capture is injective on this actual kernel. The two nullities are therefore equal and $\mathcal{A}_p$ identifies the kernels exactly.

For $j < n_p$, the image under $\mathcal{A}_p$ of the first $j+1$ eigenvectors has dimension $j+1$. Finite-dimensional min–max gives a nonzero vector u in that span such that $E_p(\mathcal{A}_p u) \ge \mu_j M_p(\mathcal{A}_p u)$. Thus

$$(1+b_p)\lambda_j \|u\|^2 \ge E_p(\mathcal{A}_p u) \ge \mu_j (1-d_p)\|u\|^2,$$

and consequently

$$\lambda_j \ge \frac{1-d_p}{1+b_p}\mu_j. \tag{35}$$

This proves the relative eigenvalue comparison. The mass statement already follows from (34).

For the stronger energy statement, (28) gives $E_p \le (1+b_p)Q_p$. Their kernels agree by the preceding argument; denote this common kernel by K_p. Both energies descend to positive forms on V_p/K_p. On this quotient define the masses

$$\overline{M}_p([x]) = \inf_{k\in K_p} M_p(x+k), \qquad \overline{N}_p([x]) = \inf_{k\in K_p} N_p(x+k). \tag{36}$$

Taking infima preserves (34). The positive generalized eigenvalues of the original pairs are exactly the generalized eigenvalues of their quotient pairs: in each class the mass minimizer is perpendicular to K_p for the corresponding mass, and the energy is constant on that class.

Put $k_p = \dim(V_p/K_p) \le N$. If $k_p = 0$, both energies vanish and there is nothing to prove. Otherwise their quotient determinants in any common basis satisfy

$$\frac{\det \overline{Q}_p}{\det \overline{E}_p} = \prod_{\lambda_j > 0} \frac{\lambda_j}{\mu_j} \frac{\det \overline{N}_p}{\det \overline{M}_p}. \tag{37}$$

In fixed Euclidean coordinates on the quotient, consider $B_p = \overline{E}_p^{-1/2} \overline{Q}_p \overline{E}_p^{-1/2}$. Every eigenvalue of B_p is at least $(1+b_p)^{-1}$. The upper eigenvalue and mass bounds imply

$$\det B_p \le c_p^{k_p} (1-d_p)^{-k_p}.$$

Dividing this product by the lower bounds for the other $k_p - 1$ eigenvalues gives

$$\lambda_{\max}(B_p) \le c_p^{k_p} (1-d_p)^{-k_p} (1+b_p)^{k_p-1}. \tag{38}$$

Both bounds tend to one uniformly for $k_p \le N$, independently of the smallest positive energy. Hence the quotient energies are relatively equivalent. Their identical kernels lift this equivalence to all of V_p, proving (32). □

Corollary 3.2. *Let $u_0, \dots, u_{n_p-1}$ be an orthonormal eigenbasis of S_p, let C_p have columns $\mathcal{A}_p u_j$, and put $\Lambda_p = \operatorname{diag}(\lambda_0, \dots, \lambda_{n_p-1})$. After enlarging e_p by a fixed factor,*

$$\left|M_p(C_p z) - |z|^2\right| \le e_p|z|^2, \qquad \left|E_p(C_p z) - z^T \Lambda_p z\right| \le e_p z^T \Lambda_p z. \tag{39}$$

Each positive-mode cross-error in the energy matrix is bounded by $e_p\sqrt{\lambda_i\lambda_j}$; every kernel row and column is exactly zero.

Proof. The definitions give $C_p^T N_p C_p = I$ and $C_p^T Q_p C_p = \Lambda_p$. Conjugate (32) by C_p. On the positive block, conjugation also by $\Lambda_p^{-1/2}$ bounds the normalized operator norm and hence every entry. Equality of kernels gives the remaining rows and columns. □

3.2. **Why the observation matters.** The conclusion is a comparison in the prescribed geometric coordinates. A near-identity change to a different orthonormal basis need not preserve relative error at the weakest scale. For example, let $E_\varepsilon = \operatorname{diag}(\varepsilon^2, \varepsilon)$ and let R_θ be a Euclidean rotation with $\theta \to 0$. Then

$$\|R_\theta^T E_\varepsilon R_\theta - E_\varepsilon\| = o(\varepsilon),$$

whereas the ratio of the first diagonal entries is $\cos^2\theta + \varepsilon^{-1}\sin^2\theta$. For $\theta = \varepsilon^{1/4}$ this ratio diverges. Rotating a zero eigendirection similarly changes the kernel in fixed coordinates. An additive estimate at the largest small eigenvalue scale therefore does not give (32).

This does not prevent a change of coordinates in the same geometric construction. For an invertible $S_p^f : V_p' \to V_p$, take

$$M_p' = (S_p^f)^* M_p, \quad E_p' = (S_p^f)^* E_p, \quad T_p' = T_p S_p^f, \quad \mathcal{A}_p' = (S_p^f)^{-1}\mathcal{A}_p.$$

All hypotheses and conclusions transform by exact congruence. There is no Euclidean condition-number assumption on S_p^f, since all the forms and maps are transported together.

In the applications, an energy may be written $E = d^*Wd$, where the adjoint uses a declared reference inner product and W is positive on the target. Its mass-normalized matrix is $M^{-1/2}d^*WdM^{-1/2}$. This elementary factorization does not construct an entire elliptic complex or determine the spectrum in another degree. Nor does Theorem 3.1 identify $Q_p(x)$ with the minimum of q_p over all sections satisfying $\mathcal{A}_p u = x$. It is the energy on the actual low spectral subspace. No exact identity $\mathcal{A}_p T_p = I$ is assumed.

4. Exact-cone trials, quotient mass and weighted lower bounds

This section fixes the central and cap geometries while their scales vary independently. The estimates are statements about the resulting surgery metrics. The comparison with the actual Ricci-flat metrics is a separate step.

4.1. **Fixed models and central representatives.** Fix an outer radius $R > 0$ and a cap parameter $A > 0$. Let g_0^R be the central metric made exactly conical on each punctured node region $0 < r < R$, and let g_i^A be the completed smoothing cap, exactly conical beyond radius $2A$. These are the fixed models of [W26a, Sections 4–5]. On the i-th cone write $g_0^R = \mathrm{d}r^2 + r^2 h_i$. Its unit-period harmonic link two-form ϑ_i is normalized by

$$\|\vartheta_i\|^2_{L^2(h_i)} = c_i = \frac{2\pi}{3}. \tag{40}$$

The link is the ordinary conifold link, diffeomorphic to $S^2 \times S^3$.

Let M_0 denote the punctured central space and identify $V = H^2(M_0;\mathbb{R})$ with $H^2(Y;\mathbb{R})$. The fixed central Hodge input is the following precise statement. Each class $D \in V$ has a central L^2 harmonic representative η_D, chosen linearly, and on every end

$$\eta_D = a_i(D)\vartheta_i + \mathrm{d}v_{i,D}, \qquad |v_{i,D}| \le C\|D\| r^{\beta-1}, \qquad |\mathrm{d}v_{i,D}| \le C\|D\| r^{\beta-2} \tag{41}$$

for a common $\beta > 0$, with the corresponding differentiated estimates. The coefficients $a_i(D)$ are the actual link restrictions $D \cdot C_i$. This is the fixed-model harmonic expansion proved in [W26a, Lemma 5.1]. Its finite-dimensional representatives and primitives are fixed before the cap scales vary.

Define the central mass and surviving subspace by

$$G_R(D,F) = \langle \eta_D, \eta_F \rangle_{g_0^R}, \qquad r(D) = (a_i(D))_i, \qquad K = \ker r, \qquad W_R = K^{\perp_{G_R}}. \tag{42}$$

The restriction of G_R to W_R is denoted M_R. The attachment calculation identifies the degree-two cohomology of every smooth surgery with K. For our family, $\dim V = 4$ and $\dim K = \dim W_R = 2$.

For arbitrary scales $\varepsilon_i > 0$ with $\ell_i = 2A\varepsilon_i < R$, glue the scaled metric $\varepsilon_i^2 g_i^A$ to the central part along its exact cone. The resulting closed smooth manifold and metric are denoted $(N_\varepsilon, g_\varepsilon)$. Set

$$L_i = \log(R/\ell_i), \qquad L_* = \min_i L_i, \qquad E_\varepsilon(D,F) = \sum_i \frac{c_i}{L_i} a_i(D) a_i(F). \tag{43}$$

Every assertion in this section is uniform as $L_* \to \infty$; no ratio of lengths is bounded. Constants may depend on the fixed R, A and central and cap models.

4.2. **Trials with exact energy.** Let χ_i vanish for $r \le \ell_i$, equal $\log(r/\ell_i)/L_i$ on $\ell_i < r < R$, and equal one for $r \ge R$. Define a global two-form $T_\varepsilon D$ to be η_D outside the ends, zero on the caps $r \le \ell_i$, and

$$T_\varepsilon D = a_i(D)\chi_i \vartheta_i + \mathrm{d}(\chi_i v_{i,D}) \tag{44}$$

on the conical annulus.

The differential here is distributional. The normal component may jump at an interface, since $\mathrm{d}\chi_i$ does. Its tangential trace is zero at the inner interface and agrees with η_D at the outer one. There is consequently no delta contribution to its exterior derivative. Equivalently the exact correction is differentiated from a continuous piecewise H^1 one-form on each collar. This gives a trial in the graph domain of d, without yet asserting membership in the graph domain of δ_ε.

Let Z_ε be the closed subspace of distributionally closed L^2 two-forms on N_ε, and let P_ε be orthogonal projection onto $Z_\varepsilon^\perp$.

Lemma 4.1. *The coexact trial $U_\varepsilon D = P_\varepsilon T_\varepsilon D$ belongs to the Hodge form domain and satisfies*

$$q_\varepsilon(U_\varepsilon D) = \|\mathrm{d}T_\varepsilon D\|^2 = \sum_i \frac{c_i |a_i(D)|^2}{L_i}, \tag{45}$$

where $q_\varepsilon(u) = \|\mathrm{d}u\|^2 + \|\delta_\varepsilon u\|^2$. Under central identification and extension by zero over missing tips,

$$\|\widetilde{T}_\varepsilon D - \eta_D\|_{L^2(M_0, g_0^R)} \le C L_*^{-1} \|D\|_{G_R}. \tag{46}$$

Proof. The exact correction contributes no exterior derivative. Thus $\mathrm{d}T_\varepsilon D = \sum_i a_i(D)\mathrm{d}\chi_i \wedge \vartheta_i$, including the interfaces just described. On a six-dimensional cone, $|\mathrm{d}r \wedge \vartheta_i|^2 = r^{-4}|\vartheta_i|^2_{h_i}$ and $\mathrm{d}V = r^5 \mathrm{d}r\,\mathrm{d}V_{h_i}$. The energy on the i-th annulus is

$$c_i|a_i(D)|^2 \int_{\ell_i}^{R} r|\chi_i'|^2\,\mathrm{d}r = \frac{c_i|a_i(D)|^2}{L_i^2}\int_{\ell_i}^{R}\frac{\mathrm{d}r}{r} = \frac{c_i|a_i(D)|^2}{L_i}.$$

The annuli are disjoint, which proves the unprojected identity.

The removed part $(I - P_\varepsilon)T_\varepsilon D$ is closed. Hence $\mathrm{d}U_\varepsilon D = \mathrm{d}T_\varepsilon D$. Orthogonality to every smooth exact two-form gives $\delta_\varepsilon U_\varepsilon D = 0$ distributionally. On a closed smooth manifold the joint graph domain of d, δ is the Hodge form domain, and $Z_\varepsilon^\perp$ is its coexact subspace. This proves (45).

On an annulus the central difference is

$$a_i(D)(\chi_i - 1)\vartheta_i + (\chi_i - 1)\mathrm{d}v_{i,D} + \mathrm{d}\chi_i \wedge v_{i,D}.$$

Its squared link-term norm, including the missing tip, is bounded by a constant times $\|D\|^2$ times

$$\ell_i^2 + L_i^{-2}\int_0^R r\log^2(R/r)\,\mathrm{d}r.$$

The remaining squared norm is bounded by a constant times $\|D\|^2$ times

$$\ell_i^{2\beta+2} + L_i^{-2}\int_0^R r^{2\beta+1}(1 + \log^2(R/r))\,\mathrm{d}r.$$

Both integrals are finite. Since $\ell_i = Re^{-L_i}$, the tip terms are also bounded by a constant times L_*^{-2}. Summing over finitely many ends proves (46). □

An end on which $a_i(D) = 0$ contributes exactly zero to the energy. This is essential for relative estimates. Cutting off the whole representative without the exact correction would introduce an unwanted additive term on such an end.

4.3. **The mass after coexact projection.** The strong convergence of $T_\varepsilon D$ does not alone control the norm of $P_\varepsilon T_\varepsilon D$. To do so we construct closed compactly supported dual forms.

Lemma 4.2. *For every sufficiently small $\rho > 0$ there is a linear map $D \mapsto \Psi_{D,\rho}$ from V to smooth compactly supported closed four-forms on M_0, vanishing on every end for $r < \rho$, such that*

$$\|\Psi_{D,\rho} - *_0\eta_D\|_{L^2(g_0^R)} \le C\rho\|D\|_{G_R}, \qquad \int_{M_0}\eta_F \wedge \Psi_{D,\rho} = G_R(F, D). \tag{47}$$

The constant is independent of ρ.

Proof. The form $\omega_D = *_0\eta_D$ is closed and has size at most $C\|D\|r^{-2}$. Write it as $\alpha_D(r) + \mathrm{d}r \wedge \beta_D(r)$ in fixed link coordinates. Then $|\alpha_D(r)|_h \le C\|D\|r^2$ and $|\beta_D(r)|_h \le C\|D\|r$. Closedness gives $\mathrm{d}_L\alpha_D = 0$, $\partial_r\alpha_D = \mathrm{d}_L\beta_D$. The tangential three-form $B_D(r) = \int_0^r \beta_D(s)\,\mathrm{d}s$ converges with its link derivatives and satisfies $\mathrm{d}B_D = \omega_D$, since $\alpha_D(0) = 0$. Its conic norm is at most $C\|D\|r^{-1}$.

Choose $\chi_\rho = 1$ for $r \le \rho$, zero for $r \ge 2\rho$, with $|\mathrm{d}\chi_\rho| \le C/\rho$, and set

$$\Psi_{D,\rho} = *_0\eta_D - \sum_i \mathrm{d}(\chi_\rho B_{i,D}).$$

This is closed and vanishes near every tip. The squared norm of its change is bounded by $C\|D\|^2 \int_0^{2\rho} r\,\mathrm{d}r$, including the cutoff-derivative term, which has size $O(\rho^{-2})$ on its shell. This proves the first estimate.

For the pairing, integrate $\eta_F \wedge \mathrm{d}(\chi_\rho B_D)$ on the space truncated at $r = \delta$. The tip boundary term is bounded by $C\|F\|\|D\|\delta^5\delta^{-2}\delta^{-1} = C\|F\|\|D\|\delta^2$. It tends to zero. All volume integrals converge, so Stokes makes the correction pairing exactly zero. □

Proposition 4.3. *For $D \in W_R$ and sufficiently large L_*,*

$$\|(I - P_\varepsilon)T_\varepsilon D\| \le C L_*^{-1}\|D\|_{G_R}. \tag{48}$$

For $D, F \in W_R$,

$$\big|\langle U_\varepsilon D, U_\varepsilon F\rangle - G_R(D,F)\big| \le C L_*^{-1}\|D\|_{G_R}\|F\|_{G_R}, \qquad q_\varepsilon(U_\varepsilon D, U_\varepsilon F) = E_\varepsilon(D,F). \tag{49}$$

Consequently, if $\nu_j(\varepsilon)$ are the generalized eigenvalues of (E_ε, M_R) on W_R, the ordered positive coexact two-form eigenvalues satisfy

$$\lambda_{2,j}^{\mathrm{coex}}(N_\varepsilon) \le (1 + C/L_*)\nu_j(\varepsilon), \qquad 1 \le j \le \dim W_R. \tag{50}$$

Proof. If $\max_i \ell_i < \rho$, extend $\Psi_{D,\rho}$ by zero onto N_ε. For a smooth closed two-form z on that space, its exterior class lies in K; call it k. Compactly supported cohomological pairing gives

$$\int_{N_\varepsilon} z \wedge \Psi_{D,\rho} = \int_{M_0} \eta_k \wedge \Psi_{D,\rho} = G_R(k, D) = 0.$$

Only the class on a common exterior containing the support is used; no smooth extension of z over the missing central tips is required. The equality extends to every closed L^2 form by heat regularization on each fixed closed smoothing. This regularization needs no estimate uniform in ε.

Since $*^2 = 1$ in degrees two and four in dimension six, $V_{\varepsilon,\rho}D = *_\varepsilon \Psi_{D,\rho}$ is perpendicular to Z_ε. Its support is in the unchanged central metric. Lemma 4.2 and (46) give

$$\|T_\varepsilon D - V_{\varepsilon,\rho}D\| \le C(L_*^{-1} + \rho)\|D\|_{G_R}.$$

Take $\rho = R/(1 + L_*)$. For large L_*, the support condition and $2\rho < R$ hold. Orthogonal projection now proves (48).

The unprojected Gram matrix converges at rate $O(L_*^{-1})$ by (46). The projected Gram matrix differs by

$$-\langle (I - P_\varepsilon)T_\varepsilon D, (I - P_\varepsilon)T_\varepsilon F\rangle,$$

which is $O(L_*^{-2})\|D\|\|F\|$. This proves the mass assertion; polarization of (45) proves the energy identity. On W_R, the energy is positive definite. The mass bound makes U_ε injective. Apply coexact min–max to the first j generalized eigendirections to obtain $\nu_j/(1 - C/L_*)$, which implies (50). □

The proof uses orthogonality to the full surviving space K. Orthogonality only to the polarizing class would not annihilate every closed smoothing form in the compactly supported pairing.

4.4. **A fixed exterior projection.** Let C be the compact central exterior with boundary $r = r_* = R/2$. It has fixed geometry and $H^2(C;\mathbb{R}) = V$. Let Z_C be the maximal distributional closed subspace of L^2 two-forms on C, imposing no tangential trace condition, and let Π_C be its orthogonal projection. Absolute harmonic representation gives a bounded class map from Z_C to V. Set

$$\mathcal{A}_\varepsilon u = \mathrm{pr}_{W_R}^{G_R}[\Pi_C(u|_C)]. \tag{51}$$

The subscript specifies the changing domain; the exterior operation itself is fixed.

Lemma 4.4. *For $u|_C$ in the maximal domain of* d,

$$\|(I-\Pi_C)u\|_{H^1(C)} \le C\|\mathrm{d}u\|_{L^2(C)}. \tag{52}$$

Proof. Put $v=(I-\Pi_C)u$. Then $\mathrm{d}v=\mathrm{d}u$. For every one-form ϕ in the maximal domain of d, its differential lies in Z_C, so $\langle v,\mathrm{d}\phi\rangle=0$. By the adjoint-domain definition, v belongs to the minimal codifferential domain and $\delta v=0$. This includes its weak zero normal trace.

The first-order absolute boundary regularity theorem gives

$$\|v\|_{H^1(C)} \le C(\|\mathrm{d}v\|+\|\delta v\|+\|v\|).$$

The weak domain and regularity used here are those of [Fre05, Definition 1.2.5, Lemma 1.2.10 and Theorem 4.3.8]. There is no imposed normal trace condition on $\mathrm{d}v$.

On this subspace also $\|v\|\le C\|\mathrm{d}v\|$. Otherwise unit vectors in $Z_C^{\perp}$ with differential tending to zero would be H^1 bounded. Rellich compactness would give a unit strong L^2 limit in both Z_C and $Z_C^{\perp}$, a contradiction. Combining the two estimates proves (52). □

4.5. **Boundary coefficients and capacity.** For a smooth two-form, let

$$B_i(u)=c_i^{-1}\int_{\{r=r_*\}} u_T\wedge *_h\vartheta_i. \tag{53}$$

This integrates the tangential trace over the whole five-dimensional boundary against a smooth three-form. It is a bounded H^1 trace functional, not an integral of an arbitrary H^1 two-form over a codimension-three cycle.

For $\zeta=\Pi_C u$, the harmonic boundary coefficient is the restriction coordinate $a_i([\zeta])$. This follows by pairing the closed tangential class with $*_h\vartheta_i$. For smooth data, first-order boundary regularity makes ζ smooth; approximation then gives the assertion in the weak setting. The fixed trace estimate and Lemma 4.4 yield

$$|B_i(u)-a_i(\mathcal{A}_\varepsilon u)| \le C\|\mathrm{d}u\|_{L^2(C)} \le C\sqrt{q_\varepsilon(u)}. \tag{54}$$

Here r vanishes on K, so taking the quotient projection has not changed the restriction coefficient.

The cap and annulus capacity bound, with its exact normalization, is

$$q_{\varepsilon,i}(u) \ge \frac{c_i|B_i(u)|^2}{\log(r_*/\varepsilon_i)+C_i}. \tag{55}$$

We recall its variational reason from [W26a, Section 4]. The closed three-form $*_h\vartheta_i$ extends into the fixed smoothing cap. Stokes and Cauchy–Schwarz bound its inner boundary coefficient by the cap d-energy. On the conical annulus the harmonic component of $\mathrm{d}u$ controls $c_i\int r|B_i'(r)|^2\,\mathrm{d}r$. Adding the two resistances gives (55). The energy of a two-form in real dimension six is invariant under constant metric scaling, so the cap constant is independent of ε_i. No special boundary condition is imposed on the competitor, and no other cap scale enters.

Proposition 4.5. *For every form in the Hodge form domain, and sufficiently large L_*,*

$$E_\varepsilon(\mathcal{A}_\varepsilon u) \le (1+CL_*^{-1/2})q_\varepsilon(u). \tag{56}$$

Moreover,

$$\begin{aligned} &\|\mathcal{A}_\varepsilon u\|_{G_R} \le C\|u\|, \qquad \mathcal{A}_\varepsilon z=0 \quad (z\in Z_\varepsilon),\\ &\|\mathcal{A}_\varepsilon U_\varepsilon D-D\|_{G_R} \le CL_*^{-1}\|D\|_{G_R} \quad (D\in W_R). \end{aligned} \tag{57}$$

Proof. Put $w_i = c_i/L_i$. The denominator in (55) is L_i plus a fixed bounded shift. Sum over the disjoint regions to obtain

$$\sum_i w_i |B_i(u)|^2 \le (1 + C/L_*) q_\varepsilon(u).$$

Equation (54), the fixed number of ends and $w_i \le C/L_*$ give

$$\sum_i w_i |B_i(u) - a_i(\mathcal{A}_\varepsilon u)|^2 \le C L_*^{-1} q_\varepsilon(u).$$

The triangle inequality in this weighted Euclidean norm proves (56), after squaring. Every boundary error is proportional to the energy of the same competitor. Smooth density extends the estimate to the form domain.

For boundedness of $\mathcal{A}_\varepsilon$, restriction and Π_C are contractions, and absolute harmonic representation and the fixed finite quotient projection are bounded. A globally closed form has exterior class in K, proving the second assertion; closed L^2 forms follow by regularization on each smoothing. Finally $\mathcal{A}_\varepsilon \eta_D = D$ on W_R and (46) gives the last estimate for $T_\varepsilon D$. Its difference from $U_\varepsilon D$ is globally closed, so both have the same observation. □

The trials and weighted observation now verify the first two hypotheses of Theorem 3.1 on the model coexact sector. The remaining mass-capture estimate concerns arbitrary low-energy forms. It requires independent-cap compactness, not merely the mass convergence of the trials proved above.

5. Independent-cap compactness and the complete model spectrum

The estimates in this section concern forms of degree one or two. The distinction matters at the middle degree: we do not assert the same weighted estimate for degree three or for arbitrary mixed-degree forms.

5.1. **A weighted estimate on one completed cap.** Let Z be a completed cap diffeomorphic to T^*S^3, with fixed smooth complete metric equal to $\mathrm{d}r^2 + r^2 h$ for $r \ge r_0$. Choose a positive smooth radius ρ_Z equal to r on the end and bounded above and below on the core. Put $\mathcal{D}_Z = \mathrm{d} + \delta$.

Proposition 5.1. *For $p = 1, 2$, every compactly supported smooth p-form v satisfies*

$$\|v\|_{L^2(Z)} \le C_p \|\rho_Z \mathcal{D}_Z v\|_{L^2(Z)}. \tag{58}$$

The constant depends only on the fixed cap.

Proof. On the cone the radial unitary transformation gives

$$U \mathcal{D} U^{-1} = J\left(\partial_r + \frac{\mathsf{A}}{r}\right), \qquad \|u\|^2 = \int \|\sigma(r)\|_{\mathcal{H}}^2 \,\mathrm{d}r, \quad \sigma = Uu. \tag{59}$$

Here $\mathcal{H}$ is the doubled space of all link forms, J is unitary, and A is self-adjoint elliptic with compact resolvent. We use the normalization in [AT15, Section 2.2, p. 1028]. For a pure p-form, the radial and tangential entries belong to the closed subspace $\mathcal{S}_p = L^2\Omega^{p-1}(L) \oplus L^2\Omega^p(L)$. It need not be invariant under A.

Set $t = \log r$, $w(t) = r^{1/2}\sigma(r)$, and $\mathsf{B} = \mathsf{A} - \frac{1}{2}$. Then

$$\|u\|^2 = \int \|w(t)\|^2 \,\mathrm{d}t, \qquad \|r\mathcal{D}u\|^2 = \int \|(\partial_t + \mathsf{B})w(t)\|^2 \,\mathrm{d}t. \tag{60}$$

We claim that

$$\mathcal{S}_p \cap \ker \mathsf{B} = \{0\} \qquad (p \neq 3). \tag{61}$$

A vector in this intersection would give a pure harmonic form $u = r^{p-3}\alpha + \mathrm{d}r \wedge r^{p-4}\beta$, of pointwise order r^{-3}. Because it has a single degree, $\mathrm{d}u$ and δu have distinct degrees and vanish separately. Their equations are

$$\mathrm{d}_L\alpha = 0, \quad (p-3)\alpha = \mathrm{d}_L\beta, \qquad \delta_L\beta = 0, \quad \delta_L\alpha + (p-3)\beta = 0.$$

It follows that $\Delta_L\beta = -(p-3)^2\beta$. Nonnegativity forces $\beta = 0$ and then $\alpha = 0$ for $p \neq 3$. Kernel vectors are smooth by ellipticity, so this proves the claim on the full kernel.

Since B has a spectral gap off its finite-dimensional kernel, (61) implies

$$\|b\| \leq C_p\|\mathsf{B}b\|, \qquad b \in \mathcal{S}_p \cap \operatorname{Dom}\mathsf{B}. \tag{62}$$

Indeed, if unit vectors in this subspace had $\mathsf{B}b_j \to 0$, their nonkernel components would tend to zero. Their kernel components have a convergent subsequence, giving a nonzero limit in both $\mathcal{S}_p$ and $\ker\mathsf{B}$. This also explains why possible mixed-degree vectors in the critical kernel cause no problem for the stated estimate.

For compactly supported w, self-adjointness and integration by parts give

$$\int \|(\partial_t + \mathsf{B})w\|^2\,\mathrm{d}t = \int \left(\|\partial_t w\|^2 + \|\mathsf{B}w\|^2\right)\mathrm{d}t.$$

Together with (60) and (62), this proves the estimate on the open cone.

Choose a cutoff equal to one far on the cone and zero on the core. Applying the cone estimate and the first-order Leibniz rule gives

$$\|v\| \leq C\|\rho_Z\mathcal{D}_Z v\| + C\|v\|_{L^2(K_Z)} \tag{63}$$

for a fixed compact set K_Z. The completed cap has no L^2 harmonic forms in degrees one and two. Indeed its exact-cone-ended metric is a scattering metric, and [HHM04, Theorem 1A] identifies its below-middle L^2 cohomology with $H^p_c(T^*S^3;\mathbb{R})$. The Thom isomorphism gives $H^p_c(T^*S^3) = H^{p-3}(S^3) = 0$ for $p = 1, 2$.

If (58) failed, unit compactly supported pure forms v_j could satisfy $\|\rho_Z\mathcal{D}_Z v_j\| \to 0$. Interior first-order elliptic estimates and Rellich compactness give a subsequence converging locally in L^2. Its limit is an L^2 harmonic pure form by Fatou and elliptic regularity, hence zero. The compact term in (63) tends to zero, contradicting the unit norm. □

5.2. **Scaling and preservation of norm.** On the i-th scaled cap put $\rho_{i,\varepsilon} = \varepsilon_i\rho_{Z_i}$; it is the physical cone radius on the end. Degreewise L^2 isometric scaling turns (58) into

$$\|v\|_{L^2(\varepsilon_i^2 g_i)} \leq C\|\rho_{i,\varepsilon}\mathcal{D}_\varepsilon v\|_{L^2(\varepsilon_i^2 g_i)}.$$

The first-order squared energy scales two powers faster than the mass, and the squared radius restores those powers.

Choose a radial cutoff χ_s equal to one on the cap and for $r \leq s$, and zero for $r \geq 2s$, with $|\mathrm{d}\chi_s| \leq C/s$. If ε_i is sufficiently small relative to s, its support is contained in the part identified with the completed cap, where $\rho_{i,\varepsilon} \leq Cs$ and $\rho_{i,\varepsilon}|\mathrm{d}\chi_s| \leq C$ on the shell. Extending $\chi_s u$ by zero and applying the estimate proves

$$\|u\|^2_{\{i\text{-th cap and } r<s\}} \leq Cs^2 q_\varepsilon(u) + C\|u\|^2_{\{s<r<2s\},i}, \qquad p = 1, 2. \tag{64}$$

The same constant works for the finitely many fixed caps. Its proof involves each scale only on its own cap.

Proposition 5.2. *Suppose every $\varepsilon_i \to 0$, the norms of pure degree-one or degree-two forms u_ε are bounded, and $q_\varepsilon(u_\varepsilon) \to 0$. A subsequence converges locally in L^2 to a smooth central L^2 closed and coclosed form u_0, and*

$$\|u_\varepsilon\|^2 - \|u_0\|^2 \longrightarrow 0. \tag{65}$$

The central restrictions extended by zero over missing tips converge strongly in central L^2, and the cap-core norms tend to zero. For finitely many sequences one can preserve the full Gram matrix on a common subsequence. For eigenforms with eigenvalues tending to zero the convergence is smooth on central compact subsets.

Proof. On fixed central compact sets the norm and first-order energy bounds give local H^1 bounds. Rellich compactness and a diagonal choice give local strong L^2 convergence. Vanishing energy gives $\mathrm{d}u_0 = \delta u_0 = 0$ distributionally. Elliptic regularity and Fatou give smoothness and finite L^2 norm.

For fixed s, apply (64) on every end and pass to the limit:

$$\limsup_{\varepsilon\to 0} \|u_\varepsilon\|^2_{\{\text{all caps and } r<s\}} \le C \sum_i \|u_0\|^2_{\{s<r<2s\},i}.$$

The right side tends to zero as $s \downarrow 0$. This and the local convergence prove strong central convergence, vanishing cap mass and (65). Simultaneous strong convergence and polarization preserve the Gram matrix of a finite family. For eigenforms, fixed-exterior elliptic estimates for the Laplacian upgrade the central convergence to smooth convergence. □

5.3. **The complete model theorem.** We use also the fixed-central Hodge statement from [W26a, Section 5]: the central L^2 closed and coclosed one-form space is zero, and the corresponding two-form space is the four-dimensional space of representatives η_D, $D \in V$. These representatives lie in the minimal graph domain. This is a fixed conic assertion; the compactness estimates above supply the independent-scale step.

Theorem 5.3. *For fixed R, A there are $L_0, \gamma > 0$ and $e(L_*) \to 0$ such that, for every scale vector with $L_* \ge L_0$, the model has the following properties.*

Its harmonic two-form space is marked by all of K and its mass converges uniformly to $G_R|_K$. There are exactly two positive two-form eigenvalues below γ, both coexact, and

$$(1 - e(L_*))\nu_j(\varepsilon) \le \lambda^+_{2,j}(\varepsilon) \le (1 + e(L_*))\nu_j(\varepsilon), \qquad j = 1, 2. \tag{66}$$

The first positive one-form and third positive two-form eigenvalues are bounded below by γ.

On the full two-dimensional positive small space S_ε, $\mathcal{A}_\varepsilon$ is an isomorphism onto W_R and

$$|M_R(\mathcal{A}_\varepsilon u, \mathcal{A}_\varepsilon v) - \langle u, v\rangle| \le e(L_*)\|u\|\|v\|, \qquad \|u - U_\varepsilon \mathcal{A}_\varepsilon u\| \le e(L_*)\|u\|. \tag{67}$$

There is no ratio restriction among the scales or positive model eigenvalues.

Proof. If the one-form gap failed, normalized eigenforms with eigenvalues tending to zero would, by Proposition 5.2, give a central L^2 harmonic one-form of norm one. This contradicts the fixed-central vanishing. Failure of a uniform threshold would allow the sequence with $L_* \ge n$ and eigenvalue below $1/n$, so the contradiction proves a uniform gap over all scale ratios.

The model has $b_2 = 2$. Proposition 4.3 gives two positive coexact eigenvalues bounded by C/L_*. If its third positive two-form eigenvalue could tend to zero, the first three positive eigenvectors together with two orthonormal harmonic forms would have vanishing energy. Proposition 5.2 would preserve five orthonormal vectors in the four-dimensional central harmonic space, which is impossible. Hence there is a uniform third-positive-eigenvalue gap.

The positive exact two-form spectrum equals the positive coexact one-form spectrum, with multiplicities, by the singular values of d. The one-form gap therefore makes both small positive two-form spaces coexact.

For $k \in K$, the trial $T_\varepsilon k$ is closed by (45) and represents the corresponding smoothing class. Its harmonic representative $h_\varepsilon(k)$ has norm at most that of the trial:

$$\|h_\varepsilon(k)\|^2 \le \|T_\varepsilon k\|^2 = G_R(k,k) + o(1).$$

Compactness gives a no-loss subsequential central limit η_D. Pair with the fixed compactly supported duals $\Psi_{F,\rho}$, for all $F \in V$ and fixed ρ. The class of the smoothing representative gives

$$\int_{N_\varepsilon} h_\varepsilon(k) \wedge \Psi_{F,\rho} = G_R(k,F).$$

Passing to the limit identifies $G_R(D,F) = G_R(k,F)$ for every F, so $D = k$. Norm preservation proves convergence of the harmonic mass. Finite-dimensional polarization and a contradiction on unit vectors make it uniform on K.

Next take normalized $u_\varepsilon \in S_\varepsilon$ along any sequence with $L_* \to \infty$. The trial upper bound gives energy at most C/L_*. A no-loss subsequence converges to η_D of norm one. For every $k \in K$, the trial $T_\varepsilon k$ is closed, so $\langle u_\varepsilon, T_\varepsilon k\rangle = 0$. Strong convergence gives $G_R(D,k) = 0$, hence $D \in W_R$. Since the observation is a bounded fixed operator on the exterior, $\mathcal{A}_\varepsilon u_\varepsilon \to D$. Therefore $M_R(\mathcal{A}_\varepsilon u_\varepsilon) \to 1$.

Every potentially violating sequence has this subsequence, so the mass convergence is uniform on all unit vectors in S_ε. Polarization gives the first estimate in (67); injectivity and dimension make the observation an isomorphism. The trial $U_\varepsilon \mathcal{A}_\varepsilon u_\varepsilon$ has the same strong central limit and vanishing cap norm. Subtracting and using a second sequential contradiction gives the reconstruction estimate.

For the span of the first j positive eigenvectors, the weighted lower bound and mass capture give

$$\frac{E_\varepsilon(\mathcal{A}_\varepsilon u)}{M_R(\mathcal{A}_\varepsilon u)} \le \frac{1 + CL_*^{-1/2}}{1 - e_0(L_*)} \frac{q_\varepsilon(u)}{\|u\|^2} \le (1 + o(1))\lambda^+_{2,j}.$$

Its image has dimension j, so generalized min–max gives the required relative lower bound. The trial upper bound proves the other side of (66). This establishes the theorem without division by the weakest eigenvalue. □

These model metrics need not be Kähler, and no complex type or primitivity conclusion has been used. Their constants may depend on R, A. The fixed-central quotient mass converges to M as $R \downarrow 0$ by the central harmonic minimization in [W26a, Section 5]. When the geometric smoothing scales satisfy $\varepsilon_i = |\tau_i|^{1/3}$ times fixed positive normalization factors, the relation

$$L_i = \frac{1}{3}|\log|\tau_i|| + O_{R,A}(1)$$

converts each energy coefficient to $(1 + o(1))\, 2\pi/|\log|\tau_i||$ multiplicatively. This conversion alone is not a comparison with an actual Ricci-flat metric.

6. Scalar estimates on Stenzel caps

The passage from the surgery metrics to the Ricci-flat metrics requires a scalar inverse which does not lose powers of the smallest cap scale. We first identify the entire low-growth harmonic space on the complete Stenzel manifold. The distinction between its pluriharmonic and quadratic directions will also determine the nonlinear construction.

6.1. **The fifteen low-growth harmonic functions.** Let

$$Z = \{z \in \mathbb{C}^4 : Q(z) = z_1^2 + \cdots + z_4^2 = 1\}, \qquad \omega = \mathrm{i}\partial\bar\partial\phi, \qquad \phi = F(s), \quad s = |z|^2,$$

carry one fixed normalization of the Stenzel metric g. Put $\rho = s^{1/3} \geq 1$ and $B_t = \{\rho < t\}$. The radial potential has the differentiated asymptotic expansion $\phi = c_0\rho^2 + o(\rho^2)$, with $c_0 > 0$. The radius ρ is comparable to the metric cone radius. Write

$$P = \Lambda_\omega \mathrm{i}\partial\bar\partial, \qquad P\phi = 3.$$

Thus P differs from the positive real scalar Laplacian by a fixed nonzero factor and sign. This convention will be used throughout the metric construction.

For $q > 6$, use the weighted norms

$$\|v\|^q_{W^{k,q}_\delta} = \sum_{j=0}^{k} \int_Z \bigl(\rho^{j-\delta}|\nabla^j v|\bigr)^q \rho^{-6}\,\mathrm{d}V_g. \tag{68}$$

The scalar Fredholm theory of [Pac13, Theorem 7.9, Proposition 7.11, Corollary 9.4 and Section 10] gives surjectivity for nonexceptional weights $\delta > -4$, index zero for $-4 < \delta < 0$, and index jumps equal to the multiplicity of each crossed indicial root. For the ordinary double point, the nonnegative scalar indicial roots strictly below three are

$$0, \quad \frac{3}{2}, \quad 2, \qquad \text{with real multiplicities } 1,\ 8,\ 6;$$

the next root is three. These are the roots of the full link spectrum in [HS17, Example 2.16, equations (2.41)–(2.42)]. In particular the kernel at every weight $2 < \delta < 3$ has dimension fifteen.

Lemma 6.1. *At a weight $2 < \delta < 3$, the real kernel of P is spanned by 1, $\operatorname{Re} z_j$, $\operatorname{Im} z_j$ for $1 \leq j \leq 4$, and*

$$q_A(z) = 2F'(s) z^* A z, \qquad A \in \mathfrak{iso}(4, \mathbb{R}). \tag{69}$$

The first nine functions have zero complex Hessian. The remaining six have degree-two leading terms and satisfy, for every $j \geq 0$,

$$\sup_Z \rho^j |\nabla^j (\mathrm{i}\partial\bar\partial q_A)|_g \leq C_j |A|. \tag{70}$$

Proof. The first nine functions are pluriharmonic. If $A \in \mathfrak{iso}(4, \mathbb{R})$, then A is Hermitian, $A^{\mathsf{T}} = -A$, and $G_t = e^{tA}$ preserves Q and its residue volume form. Pulling back the Stenzel Calabi–Yau identity by G_t and differentiating gives

$$3\mathrm{i}\partial\bar\partial \left.\frac{\mathrm{d}}{\mathrm{d}t}\right|_{t=0} \phi(G_t z) \wedge \omega^2 = 0.$$

The derivative is (69), so $Pq_A = 0$. The radial Stenzel formula and its differentiated asymptotics, as in [CH13, Lemma 5.14], give the leading term

$$\frac{4c_0}{3} s^{-1/3} z^* A z.$$

This map of A is injective. Indeed write $A = \mathrm{i}B$ with B real skew-symmetric and evaluate z^*Az on the null vectors $z = e_j + \mathrm{i}e_k$: these values recover, up to a fixed nonzero factor, every entry B_{jk}. The six leading terms are therefore independent. Together with the lower-degree functions they exhaust the fifteen-dimensional Fredholm kernel. Smoothness on the fixed core and the differentiated expansion on the end prove (70). □

Choose a basis $H_1, \dots, H_{15}$ as in the lemma and denote its growth degrees by $d_a \in \{0, 3/2, 2\}$. For all sufficiently large T,

$$\sum_a |b_a| T^{d_a} \le C \left\| \sum_a b_a H_a \right\|_{L^\infty(B_{2T} \setminus B_T)}. \tag{71}$$

To prove this, dilate the annulus to a fixed cone annulus. The normalized functions $T^{-d_a} H_a$ converge there to independent homogeneous functions: within each degree their link terms are independent, and distinct degrees have different radial powers. Their limiting L^2 Gram matrix is positive definite. Its least eigenvalue stays positive for all large T, which proves the estimate.

6.2. **A local complex-Hessian estimate.** For $\varepsilon > 0$ put $g_\varepsilon = \varepsilon^2 g$, $r_\varepsilon = \varepsilon \rho$, and $P_\varepsilon = \varepsilon^{-2} P$. All derivatives in the following statement use g_ε.

Proposition 6.2. *For every $\mu > 0$ and integer $m \ge 0$ there are $C, T_0 > 0$ such that the following holds. Suppose $T = R/\varepsilon \ge T_0$, $P_\varepsilon u = f$ on B_{4T}, and $c \in \mathbb{R}$. Set*

$$K = \max_{0 \le j \le m+1} \sup_{B_{4T}} r_\varepsilon^{j-\mu} |\nabla^j (f - c)|.$$

Then

$$\max_{0 \le j \le m} \sup_{B_T} r_\varepsilon^j |\nabla^j (\mathrm{i}\partial\bar\partial u)| \le C \left(R^{-2} \|u\|_{L^\infty(B_{4T})} + |c| + R^\mu K \right). \tag{72}$$

The constants are independent of ε and of the phase in a holomorphic identification of a smoothing quadric with Z.

Proof. Choose $0 < \eta < \min(\mu, 1)$ and put $\delta = 2 + \eta$. The fixed-cap Fredholm theory gives a bounded right inverse

$$G_\delta : W^{m+1,q}_{\delta-2} \longrightarrow W^{m+3,q}_{\delta}, \qquad P G_\delta = \mathrm{Id}.$$

Weighted Sobolev embedding gives

$$\sup_Z \rho^{j-\delta} |\nabla^j G_\delta F| \le C \|F\|_{W^{m+1,q}_{\delta-2}}, \qquad j \le m+2. \tag{73}$$

First take $c = 0$. On the unit cap the equation is $Pu = \varepsilon^2 f$, and $|\nabla_g^j f| \le K \varepsilon^\mu \rho^{\mu-j}$. Choose χ_T equal to one on B_{2T} and zero outside B_{3T}, with the usual scaled derivative bounds. Put

$$U = \|u\|_{L^\infty(B_{4T})}, \qquad \mathcal{B} = U + R^{2+\mu} K.$$

Interior estimates on the transition annulus bound the commutator $[P, \chi_T] u$ in $W^{m+1,q}_{\delta-2}$ by $C T^{-\delta} \mathcal{B}$. The interior source has the same bound, since

$$\|\chi_T \varepsilon^2 f\|_{W^{m+1,q}_{\delta-2}} \le C K \varepsilon^{2+\mu} T^{\mu-\eta} = C T^{-\delta} R^{2+\mu} K.$$

Here the radial integral is bounded by $C \int_1^{3T} \rho^{q(\mu-\eta)} \, \mathrm{d}\rho/\rho$; the strict inequality $\eta < \mu$ avoids a logarithmic loss.

Define $v = G_\delta P(\chi_T u)$ and $H = \chi_T u - v$. The latter belongs to the complete-cap harmonic kernel. On $B_{2T} \setminus B_T$, (73) bounds v and H by $C\mathcal{B}$. Equation (71) bounds the six quadratic coefficients of H by $C T^{-2} \mathcal{B}$; the other nine modes have zero complex Hessian. Thus

$$\sup_{B_T} \rho^j |\nabla_g^j (\mathrm{i}\partial\bar\partial H)| \le C T^{-2} \mathcal{B}, \qquad j \le m.$$

For v, (73) gives the bound $CT^{-\delta}\mathcal{B}\rho^{\delta-2} \le CT^{-2}\mathcal{B}$. Since $u = H + v$ on B_T, this proves the unit-metric estimate, including the core. Under homothety its left side gains ε^{-2}, and $\varepsilon^{-2}T^{-2} = R^{-2}$. This proves (72) for $c = 0$.

For general c, subtract $(c/3)\varepsilon^2\phi$. Its P_ε-image is c, its supremum is at most $C|c|R^2$, and its complex Hessian is the parallel tensor $(c/3)\omega_\varepsilon$. This gives the remaining term. A phase change is a scalar unitary coordinate change, so the constants are unchanged. □

The positive order of the nonconstant forcing is essential to this argument. On the cone, if Y is a root-two link eigenfunction, then $P(r^2 \log r\, Y)$ is a nonzero constant multiple of Y. Thus a bounded angular source can produce a logarithmic complex Hessian. In particular, a nonlinear proof cannot discard the quadratic directions on the ground that their Hessians are bounded. We will incorporate their exact automorphism family.

7. Initial Kähler metrics on the full smoothing base

We use the polarized family $(X_p, \mathcal{L}_p, \Omega_p)$ of Section 2. Let $\omega_0 \in c_1(\mathcal{L}_0)$ be its central singular Ricci-flat metric. The central conical theorem of [HS17, Theorem 1.4 and Lemma A.1] applies: the nodes have strongly regular Calabi–Yau cones, and a full-support line in the period base gives a polarized smoothing over a disc. It supplies fixed holomorphic conical charts with potentials

$$\phi_{0,i} = \phi_{C,i} + v_i, \qquad \phi_{C,i} = c_i^0 |z|^{4/3}, \qquad |\nabla_C^j v_i| \le C_j r_i^{2+\mu-j} \tag{74}$$

for some $\mu > 0$. Here r_i is the fixed cone radius, comparable to $|z|^{2/3}$. We decrease μ so that $\mu \le 3/2$. The constant c_i^0 in the potential is unrelated to the link norm c_i in the capacity energy.

Proposition 7.1. *For every sufficiently small fixed $R > 0$ and all sufficiently small $p \in B^\circ$, there is a smooth Kähler form $\widetilde{\omega}_p^R \in c_1(\mathcal{L}_p)$ with the following properties.*

On $r_i < R$ it is a normalized Stenzel form $\sigma_{i,p}$, of length scale

$$\varepsilon_i = b_i(p)^{1/2} |\tau_i(p)|^{1/3}, \qquad b_i(p) \longrightarrow b_i(0) > 0. \tag{75}$$

Outside $r_i < 2R$ it converges smoothly to ω_0 under exterior family identifications.

Let Θ_p be the positive density associated with $\Omega_p \wedge \overline{\Omega_p}$, with the fixed conventional complex factor, and put

$$V = \int_{X_p} c_1(\mathcal{L}_p)^3, \quad I(p) = \int_{X_p} \Theta_p, \quad \kappa(p) = V/I(p), \quad F_p^R = \log \frac{\kappa(p)\Theta_p}{(\widetilde{\omega}_p^R)^3}.$$

Then $I(p) \to I(0) > 0$. There is a smooth radius function ϱ_p, comparable to the scaled Stenzel radius on the node regions and to one on the fixed exterior, such that, for every fixed k,

$$\limsup_{\substack{p \to 0 \\ p \in B^\circ}} \max_{j \le k} \sup_{X_p} \varrho_p^j |\nabla^j F_p^R| \le C_k R^\mu. \tag{76}$$

On each Stenzel region the stronger identity and estimate hold:

$$F_p^R = \log \left| \frac{a_i(z,p)}{a_i(0,p)} \right|^2, \qquad |\nabla^j F_p^R| \le C_j \varrho_p^{3/2-j}, \qquad \Omega_p = a_i(z,p)\Omega_{Q,\tau_i}. \tag{77}$$

The derivatives use the initial metric. The constants are independent of the ratios and phases of the node parameters. The parameter neighbourhood can depend on R and k.

7.1. **Family charts and Hermitian weights.** The conical charts in (74) may be chosen compatibly with nearby quadric fibres. Indeed lift the fixed central transition to an invertible ambient holomorphic germ Φ. Since it preserves the node, $Q\circ\Phi = uQ$ for a holomorphic unit u. After choosing a square root, the map $\Psi = u^{-1/2}\Phi$ satisfies $Q\circ\Psi = Q$ and is invertible. It therefore acts on all nearby fibres. The nonconstant scalar factor changes the central metric at relative order $O(|z|) = O(r_i^{3/2})$; its constant part changes only the fixed normalization and phase. Thus (74) holds in these family charts after decreasing μ if needed.

The central form is the normalized Chern form of a Hermitian metric h_0 on $\mathcal{L}_0$ over X_0^{reg}: use the bounded-potential Calabi–Yau solution relative to a smooth background Chern form. We normalize Hermitian weights so that their Chern form is $\mathrm{i}\partial\bar\partial\phi$ and represents c_1; the factor 2π is included in the weight convention.

Choose a holomorphic frame of $\mathcal{L}$ on each node neighbourhood. Its h_0-weight can be made exactly $\phi_{0,i}$. The difference is pluriharmonic on the punctured node. Since its link is $S^2\times S^3$, its first real cohomology vanishes; the difference is the real part of a single-valued holomorphic function. Hartogs extension on the normal threefold node extends that function across the vertex. Lift its germ to the ambient chart and extend the resulting frame change to the family.

For fixed R, the complement of the radius-$R/2$ node regions is a compact smooth part of the family. Extend h_0 smoothly there, for example by extending its logarithm relative to a background metric in a smooth tubular neighbourhood and using a partition of unity. Denote the extension by $h_{\mathrm{ext},p}$. On each fixed collar its weight $\phi_{\mathrm{ext},i,p}$ tends to $\phi_{0,i}$ in every C^k norm. No rate uniform as $R\downarrow 0$ is required.

7.2. **Volume and the exact local normalization.** Choose a holomorphic nonvanishing ambient representative for the coefficient a_i in (77). Its value at $(0,p)$ is taken in this chosen representative, even though $z=0$ does not belong to the smooth quadric.

We first prove continuity of $I(p)$ without restricting the relative scales. Under $z=\sqrt{\tau}\,y$, the residue form pulls back to $\tau\Omega_{Q,1}$ and the Stenzel metric scales by $|\tau|^{2/3}$. The unit Stenzel volume has Euclidean growth in real dimension six. Consequently the $\Theta_{Q,\tau}$-volume of $r_i<s$ is at most Cs^6, uniformly for small τ. If the region meets the cap core, the same estimate follows from the unit-cap volume bound for radii above its positive minimum. Boundedness of a_i preserves this estimate. On the complement of these balls the densities converge smoothly. Letting s tend to zero proves $I(p)\to I(0)$ along every sequence in the base. The limit is positive on the fixed exterior.

The intersection number V is constant in the flat polarized family, so $\kappa(p)\to\kappa(0)>0$. Normalize the holomorphic scaling of the unit Stenzel form by $\omega_{\mathrm{St},\tau}^3 = \kappa_{\mathrm{St}}\Theta_{Q,\tau}$ and set

$$b_i(p) = \left(\frac{\kappa(p)|a_i(0,p)|^2}{\kappa_{\mathrm{St}}}\right)^{1/3}, \qquad \sigma_{i,p} = b_i(p)\omega_{\mathrm{St},\tau_i(p)}. \tag{78}$$

The limiting cone form of $\sigma_{i,p}$ is exactly $\omega_{C,i}$: both have the standard quadric shape in the chosen conical chart, and their volume coefficient is $\kappa(0)|a_i(0,0)|^2$. That coefficient fixes the positive metric scale. This proves (75). The Stenzel potentials $\psi_{i,p}$ can be normalized by constants so that they converge to $\phi_{C,i}$ with all derivatives on each fixed punctured annulus.

7.3. **Gluing in the prescribed polarization.** Choose $\chi_i=0$ on $r_i\le R$ and $\chi_i=1$ on $r_i\ge 2R$, with $|\nabla^j\chi_i|\le C_jR^{-j}$. In the node frame use the Hermitian weight

$$\widetilde{\phi}_{i,p} = \psi_{i,p} + \chi_i(\phi_{\mathrm{ext},i,p} - \psi_{i,p}). \tag{79}$$

It agrees with the exterior weight near the outer overlap. These weights define a global Hermitian metric on the original line bundle $\mathcal{L}_p$. Its normalized Chern form therefore represents exactly $c_1(\mathcal{L}_p)$.

The form is positive on the core because it is Stenzel, and on the fixed exterior by convergence to ω_0. On the transition annulus the limit at fixed R is $\mathrm{i}\partial\bar\partial(\phi_{C,i}+\chi_i v_i)$. Every differentiated cutoff term has relative size $O(R^\mu)$ by (74). For R small the limiting form is positive, and for all sufficiently small p positivity follows by smooth convergence on this fixed annulus.

On the core, (78) makes the logarithmic residual exactly (77). A holomorphic logarithm exists on a smaller ambient neighbourhood. Its Taylor expansion has no z-constant term. A linear coordinate has conic degree $3/2$; its differentiated estimates are uniform on every scaled cap, including its core. The higher Taylor terms have higher degree, and the convergent series gives the stated estimate at every fixed order. On the transition annulus the metric error is $O(R^\mu)$ with scaled derivatives, and the central volume coefficient varies by $O(R^{3/2})$. On the fixed exterior the residual tends smoothly to zero. These estimates prove (76) and complete the proof of Proposition 7.1. Notice the exact compatibility

$$\int_{X_p} e^{F_p^R}(\widetilde{\omega}_p^R)^3 = V. \tag{80}$$

7.4. **Comparison with the surgery metrics.** The initial metrics satisfy

$$\limsup_{p\to 0}\sup|\Phi_p^*\widetilde{g}_p^R-\bar g^{R,A}_{\varepsilon(p)}|_{\bar g^{R,A}_{\varepsilon(p)}}\le CR^\mu+Ce_{\mathrm{AC}}(A),\qquad e_{\mathrm{AC}}(A)\longrightarrow 0. \tag{81}$$

Here the model metrics and their fixed central and cap modifications are those of Sections 4 and 5. On the fixed exterior the initial metric tends to g_0, and its difference from the central model is $O(R^\mu)$. On each inner region it is exactly normalized Stenzel. The completed cap differs from that metric only by the fixed large-radius asymptotically conical error $e_{\mathrm{AC}}(A)$. The same bound covers the intervening annuli.

The identifications can also accommodate all phases. The normal graph map between $Q=\tau$ and $Q=0$ tends to the identity away from the core; identifying the core with $Q=1$ uses a choice of $\sqrt{\tau}$. The two cone-end maps can differ by a bounded Reeb rotation, an isometry of the cone. Interpolate that rotation in $\log r$ between $2A\varepsilon_i$ and R, keeping it constant near the endpoints. Its radial angular derivative in the unit frame is at most $C/\log(R/(2A\varepsilon_i))$. Thus its relative metric error tends to zero as that neck length tends to infinity. The finite maximum tends to zero without a ratio bound. The maps are chosen fibrewise; a single-valued trivialization around the discriminant is not asserted. This proves (81) on the entire manifold, in the order: fixed R, A, then $p\to 0$.

The construction parallels the Hermitian-weight method in [Ben26, Proposition 5.9]; the uniform estimates here use the full period base. Neither (76) nor (81) yet compares the actual Ricci-flat metric with the surgery model.

8. A global scalar inverse with all low cap modes

Fix $k\ge 2$, $0<\alpha<1$, and $0<\eta<\min(\mu,1)$, where μ is the rate in Section 7. Put $\delta=2+\eta$. Choose a small fixed radius r_0 with disjoint node neighbourhoods of radius $6r_0$, and take $R<r_0/10$. For fixed R all cap cores lie inside $r_i<R$ once p is sufficiently small. Write

$$P_p^R=\Lambda_{\widetilde{\omega}_p^R}\mathrm{i}\partial\bar\partial.$$

Weighted Hölder norms $C^{j,\alpha}_{\gamma}$ use the derivative weights $\varrho_p^{l-\gamma}$ and the Hölder seminorm on balls of radius $c\varrho_p$, with weight $\varrho_p^{j+\alpha-\gamma}$. All charts are rescaled by their radius; the fixed exterior uses ordinary Hölder norms.

8.1. **The domain and its normalization.** The explicit construction of the initial metric gives

$$\|\widetilde{g}_p^R - \sigma_{i,p}\|_{C^{k+2,\alpha}_{\mu}(\{r_i<5r_0\},\sigma_{i,p})} \le C. \tag{82}$$

Here a tensor of weight μ has pointwise size $O(\varrho_p^{\mu})$ in the indicated metric. This is a differentiated comparison in the actual holomorphic node chart. It follows directly on the core, where the difference is zero. On the radius-R transition, differentiation of (79) gives the bound from the order $2+\mu$ central potential. On $2R < r_i < 5r_0$, use the central conical expansion and smooth convergence at fixed R. The latter errors can be made at most one in this weighted norm by shrinking the parameter neighbourhood. Higher derivative bounds give its Hölder part. The constant in (82) is independent of R, p; the permitted neighbourhood depends on R.

Choose a nonnegative smooth density supported in a fixed compact subset of the unit cap and positive on an open set. The Gram matrix of the fifteen functions H_a against this density is nonsingular: a vanishing combination on that open set vanishes everywhere by unique continuation, and its independent leading terms then have zero coefficients. Let ℓ_a be its dual moment functionals, so that $\ell_a(H_b) = \delta_{ab}$.

On the i-th scaled cap define

$$H_{i,a,p} = \varepsilon_i^{d_a} H_a, \qquad b_{i,a}(u) = \varepsilon_i^{-d_a} \ell_a(u|_i). \tag{83}$$

The right side is read after pullback to the unit cap. Fix cutoffs $\chi_i = 1$ for $r_i < r_0$ and $\chi_i = 0$ for $r_i > 2r_0$. Their supports are disjoint, and they are one on the rescaled moment supports. Put

$$v(u) = u - \sum_{i,a} b_{i,a}(u)\chi_i H_{i,a,p}, \qquad \|u\|_{X_p^R} = \|v(u)\|_{C^{k+2,\alpha}_{2+\eta}} + \sum_{i,a} |b_{i,a}(u)|. \tag{84}$$

Every core moment of $v(u)$ is zero. Thus (84) is a unique decomposition, equivalently a product of the closed zero-moment remainder space and the 120 real coefficients. For each fixed smooth fibre it is an equivalent norm on the ordinary Hölder space; uniform equivalence with the unweighted norm is not claimed.

Choose a fixed exterior probability density whose support misses the cutoffs and transport it by the exterior identifications. Let $Q(u)$ be its integral against u. Define

$$X^R_{p,0} = \{u \in X^R_p : Q(u) = 0\},$$
$$Y^R_p = C^{k,\alpha}_{\eta}, \qquad Y^R_{p,0} = \left\{ f \in Y^R_p : \int_{X_p} f \, \mathrm{d}V_{\widetilde{g}_p^R} = 0 \right\}.$$

The domain normalization and range compatibility use different densities.

Theorem 8.1. *There are $R_1 > 0$ and C such that, for each $0 < R < R_1$ and all sufficiently small $p \in B^\circ$,*

$$P_p^R : X^R_{p,0} \longrightarrow Y^R_{p,0} \quad \textit{is an isomorphism,} \qquad \|(P_p^R)^{-1} f\|_{X_p^R} \le C\|f\|_{Y_p^R}. \tag{85}$$

Its forward norm is also uniformly bounded. Moreover,

$$\|u\|_{\infty} + \max_{j\le k} \sup \varrho_p^j |\nabla^j(\mathrm{i}\partial\bar\partial u)| \le C\|u\|_{X_p^R}. \tag{86}$$

The constants depend on the fixed r_0, η, k, α and central geometry, but not on R or the relative cap scales.

8.2. **A negative-weight inverse and a global supremum bound.** Fix $-4 < \beta < 0$ and $q > 6$. Pacini's desingularization theorem [Pac13, Definition 11.6 and Theorem 12.3(1)] applies to the original fixed central conical metric and the eight separate complete Stenzel caps. Its models m_ε match at $r_i \asymp \varepsilon_i^\theta$ for one fixed $0 < \theta < 1$. Independent scales are allowed because the caps are separate connected components before gluing. The central space is connected, so the kernel consists of one global constant. Its complement can be chosen as $\ker Q$: the extension cutoff equals one on the fixed support of Q. The resulting negative-weight inverse constant is fixed before R is chosen.

We compare these models with the initial metric through scaled derivatives:

$$\max_{j\le k+2} \sup \varrho_p^j |\nabla_{m_\varepsilon}^j (\Phi_p^* \widetilde{g}_p^R - m_\varepsilon)| \le C R^\mu + e_R(p), \qquad e_R(p) \longrightarrow 0. \tag{87}$$

For fixed R, the gluing radii lie inside R. There the initial metric is Stenzel; at the model transition the conically singular and asymptotically conical errors tend to zero separately with each ε_i. Between that transition and R, the difference from the central metric is bounded by CR^μ and the cap tail. The initial transition and the exterior obey the same estimate. For phases choose $\theta < a < 1$ and interpolate any bounded Reeb rotation between ε_i^a and ε_i^θ. Every scaled derivative of a smooth logarithmic interpolation is $O(1/|\log \varepsilon_i|)$. Both endpoints are asymptotically conical, since $\varepsilon_i^a/\varepsilon_i \to \infty$. This proves (87), including phases and differentiated operator coefficients, without a ratio assumption.

For the possibly non-Kähler model let P_m be the same fixed multiple of the real scalar Laplacian as P_p^R on the Kähler fibre. The augmented model map

$$(u, c) \longmapsto P_m u + c, \qquad Q(u) = 0,$$

is a uniform isomorphism from $(W_\beta^{k+2,q} \cap \ker Q) \oplus \mathbb{R}$ to $W_{\beta-2}^{k,q}$. Indeed c is the model-volume mean of the target. Its norm is uniformly bounded by Hölder's inequality in the measure $\varrho^{-6}\mathrm{d}V$: the required radial power is $\varrho^{q'(\beta+4)}$, which is integrable since $\beta > -4$. The total volume has a positive exterior lower bound, and constant functions have uniformly bounded domain and target norms at these negative weights. Pacini's normalized inverse handles the remaining mean-zero target.

By (87), the two augmented maps differ by a small operator norm, first for small fixed R and then small p. Neumann inversion gives the corresponding bound for the initial metric. If f has zero current volume mean, integrating $P_p^R u + c = f$ gives $c = 0$. Consequently

$$\|u\|_{W_\beta^{k+2,q}} \le C \|f\|_{W_{\beta-2}^{k,q}} \le C \|f\|_{Y_p^R}, \qquad Q(u) = 0. \tag{88}$$

The second inequality uses the positive radial power $\eta - \beta + 2$ and has no logarithmic loss. Weighted embedding bounds u on the complement of the radius-r_0 regions.

To control the cores, use the exact Stenzel potential $\phi_{i,p} = \varepsilon_i^2 \phi$ as a barrier. It is bounded by Cr_0^2 there, and (82), with r_0 small, gives

$$P_p^R \phi_{i,p} = \mathrm{tr}_{\widetilde{\omega}_p^R} \sigma_{i,p} \ge c_* > 0.$$

Apply the maximum principle to $u \pm c_*^{-1} \|f\|_\infty \phi_{i,p}$ with boundary at radius r_0. This gives

$$\|u\|_\infty \le C(r_0) \|f\|_{Y_p^R}. \tag{89}$$

Ordinary interior Schauder estimates now control all required norms on fixed-radius annuli.

8.3. **The positive-weight cap expansion.** On the fixed complete Stenzel cap, [CH13, Theorems 2.9 and 2.11] gives a surjective Fredholm map $C^{k+2,\alpha}_{2+\eta} \to C^{k,\alpha}_{\eta}$. Its kernel is Lemma 6.1. Subtract the core moments from a bounded right inverse to obtain G_δ with $\ell_a(G_\delta F) = 0$.

For completeness, the Hölder seminorm printed in that reference compares pairs below a fixed convexity radius. Our scaled input norm controls that norm, so the inverse first gives the weighted supremum estimate for its output. Schauder estimates on balls comparable to ρ then give the full scaled Hölder estimate, with the compact core treated by fixed balls. Thus this inverse is bounded in precisely our norms, with no loss in η. Under homothety its physical norm remains uniform because $\delta = \eta + 2$.

Choose $\zeta = 1$ for $r_i < 3r_0$ and $\zeta = 0$ for $r_i > 4r_0$. On the complete scaled cap set

$$w = G_{\delta,i,p} P_{\sigma_{i,p}}(\zeta u), \qquad H = \zeta u - w = \sum_a b_a H_{i,a,p}. \tag{90}$$

Since w has zero core moments, these b_a are exactly (83). Put $V_i = \|w\|_{C^{k+2,\alpha}_\delta}$ and $U = \|u\|_\infty$. The annular Gram estimate (71), now at physical radius r_0, gives

$$|b_{(2)}| \le C(r_0^{-2} U + r_0^{\eta} V_i) \tag{91}$$

for the six degree-two coefficients.

On $r_i < 3r_0$, the source in (90) is $f + (P_\sigma - P_p^R)u$, and $u = H + w$. The constant and coordinate components of H have identically zero complex Hessian for both Kähler metrics. For the other components, (82) yields

$$\|(P_\sigma - P_p^R)H\|_{C^{k,\alpha}_\eta} \le C r_0^{\mu-\eta} |b_{(2)}|,$$
$$\|(P_\sigma - P_p^R)w\|_{C^{k,\alpha}_\eta} \le C r_0^{\mu} V_i.$$

The use of the complex Hessian is essential here; the real Hessians of the pluriharmonic coordinate functions do not vanish. On the cutoff annulus, (89) and ordinary Schauder estimates instead bound all source and commutator terms. The cap inverse therefore gives

$$V_i \le C \left(\|f\|_Y + r_0^{-\delta} U + r_0^{\mu} V_i + r_0^{\mu-\eta} |b_{(2)}| \right).$$

Insert (91) and fix r_0 small enough to absorb the resulting $C r_0^{\mu} V_i$. Equation (89) gives $V_i \le C(r_0)\|f\|_Y$. The same annular Gram estimate then bounds all fifteen coefficients, including degrees zero and 3/2. On $r_i < r_0$ the canonical remainder $v(u)$ equals w. On the fixed cutoff annuli and exterior its norm is controlled by the coefficient bounds and ordinary Schauder estimates. This proves the inverse estimate in (85).

8.4. **Isomorphism and the tensor estimate.** For each fixed smooth compact fibre, the scalar equation has a solution for every zero-volume-mean Hölder target, unique up to a constant. Impose $Q(u) = 0$. The preceding estimates apply directly to these Hölder solutions by elliptic theory; no assertion of density of smooth functions in the full Hölder norm is needed.

The forward map is also bounded. The remainder has weight $2+\eta$, so its image has weight η. On a node chart the pluriharmonic modes have zero image, and the quadratic modes have image $O(\varrho_p^{\mu})$ by (82). Cutoff derivatives occur at fixed radius. Integration gives the required range condition, and the only global harmonic functions are constants, removed by Q.

Finally the remainder has complex Hessian $O(\varrho_p^{\eta})$; each quadratic mode has bounded complex Hessian with scaled derivatives; and each pluriharmonic mode has zero such Hessian

where its cutoff is constant. All other terms occur at fixed radius. Their supremum norms are bounded as well. This proves (86) and completes the proof of Theorem 8.1.

9. Uniform correction to the actual Ricci-flat metrics

We now solve the Monge–Ampère equation on the prescribed fibres in their prescribed polarization. The six quadratic harmonic directions at each node are represented by finite automorphisms, so that their nonlinear volume contribution cancels exactly.

Theorem 9.1. *Let ω_p be the Ricci-flat Kähler form in $c_1(\mathcal{L}_p)$. With the fixed choices of Section 8, there are $R_1 > 0$ and C_k such that, for each fixed $0 < R < R_1$ and sufficiently small $p \in B^\circ$,*

$$\omega_p = \widetilde{\omega}_p^R + \mathrm{i}\partial\bar\partial u_p^R, \qquad Q(u_p^R) = 0,$$

where u_p^R is real and

$$\limsup_{\substack{p\to 0\\ p\in B^\circ}} \max_{j\le k} \sup_{X_p} \varrho_p^j |\nabla^j(\mathrm{i}\partial\bar\partial u_p^R)|_{\widetilde{g}_p^R} \le C_k R^{\mu-\eta}. \tag{92}$$

Consequently the compatible surgery identifications satisfy

$$\limsup_{\substack{p\to 0\\ p\in B^\circ}} \sup |\Phi_p^* g_p - \bar g^{R,A}_{\varepsilon(p)}|_{\bar g^{R,A}_{\varepsilon(p)}} \le CR^{\mu-\eta} + CR^{\mu} + Ce_{\mathrm{AC}}(A). \tag{93}$$

There is no restriction on scale ratios or phases. The limits are taken with R, A fixed before $p \to 0$, and then $R \downarrow 0$, $A \to \infty$. No parameter-derivative estimate or arbitrary simultaneous choice $R = R(p)$ is asserted.

9.1. **The nonlinear potential coordinates.** Use the moment decomposition of (84). Separate the nine pluriharmonic coefficients at each node from the six quadratic coefficients, and identify the latter with $A_i \in \mathfrak{iso}(4,\mathbb{R})$ using the fixed basis in Lemma 6.1. For the Stenzel potential $\psi_{i,p}$ define

$$\Psi_{i,p}(A_i)(z) = \psi_{i,p}(e^{A_i} z) - \psi_{i,p}(z).$$

After shrinking the fixed charts this is defined for A_i in a fixed small ball. Let E_p^R be the product of the Q-zero, zero-core-moment remainder space and the finite coefficients $b = (b_{i,a})_{a\le 9}$, $A = (A_i)$, with norm

$$\|x\|_E = \|v\|_{C^{k+2,\alpha}_{2+\eta}} + \sum_{i,a} |b_{i,a}| + \sum_i |A_i|.$$

For $x = (v, b, A)$ put

$$S_p(x) = v + \sum_i \chi_i \left(\sum_{a=1}^{9} b_{i,a} H_{i,a,p} + \Psi_{i,p}(A_i) \right). \tag{94}$$

The derivative $J_p = DS_p(0)$ is the canonical linear identification with $X^R_{p,0}$ and has uniformly bounded inverse by the norm definition. Also $Q(S_p(x)) = 0$.

For small A, the differentiated radial Stenzel formula gives

$$\begin{aligned} |\nabla^j \Psi_{i,p}(A)| &\le C_j |A| \varrho_p^{2-j}, \\ |\nabla^j(\sigma_{i,p,A} - \sigma_{i,p})| &\le C_j |A| \varrho_p^{-j}, \qquad \sigma_{i,p,A} = (e^A)^* \sigma_{i,p}. \end{aligned} \tag{95}$$

Positive-order derivatives in A have the same radius weights, with products of direction norms in place of $|A|$. The estimates include scaled Hölder norms. On the unit cap they follow from

smoothness on its core and the differentiated asymptotic expansion on its end; the matrices e^A and their inverses range in a compact set, so radii and scaled derivatives are comparable. Homothety proves uniformity in every ε_i.

The exact identity

$$\sigma^3_{i,p,A} = \sigma^3_{i,p} \tag{96}$$

is decisive. The matrix e^A preserves the quadric and residue form because $A^{\mathsf{T}} = -A$ and $\operatorname{tr} A = 0$. Pulling back the normalized Stenzel Calabi–Yau equation proves (96) for finite A.

We do not assert that S_p maps a uniform parameter ball into a uniform ball of the linear norm X_p^R. Its nonlinear degree-two angular profile need not have that property. Instead we estimate the volume operator directly on E_p^R. The variable remainder v is not pulled back by e^A, so no differentiability loss from composition of variable Hölder functions is involved.

9.2. **Uniform weighted nonlinear estimates.** Define

$$\mathcal{M}_p^R(x) = \log \frac{(\widetilde{\omega}_p^R + \mathrm{i}\partial\bar\partial S_p(x))^3}{(\widetilde{\omega}_p^R)^3}. \tag{97}$$

On a fixed sufficiently small ball in E_p^R the perturbed form is positive, $\mathcal{M}_p^R(x) \in Y_p^R$, and

$$\|D\mathcal{M}_p^R(x) - D\mathcal{M}_p^R(y)\|_{E\to Y} \le C\|x-y\|_E. \tag{98}$$

Here and below the radius and constants are independent of R, p.

We verify the weight near a node. Write $e_i = \widetilde{\omega}_p^R - \sigma_{i,p}$; (82) gives $e_i = O(\varrho_p^\mu)$ with scaled derivatives. On $r_i < r_0$ the cutoffs are constant and the pluriharmonic terms have zero complex Hessian. The perturbed form is therefore

$$\sigma_{i,p,A_i} + e_i + B, \qquad B = \mathrm{i}\partial\bar\partial v = O(\|v\|_E \varrho_p^\eta).$$

By (96), its logarithmic volume ratio is exactly

$$\log\det\bigl(I + \sigma^{-1}_{i,p,A_i}(e_i + B)\bigr) - \log\det\bigl(I + \sigma^{-1}_{i,p} e_i\bigr). \tag{99}$$

For small fixed r_0 and parameter radius, these matrices remain in a fixed positive cone. Inversion and logarithmic determinant have bounded derivatives there. Each variation in v has weight η. A variation in A differentiates σ_A^{-1} but retains the factor $e_i + B$; all pure A derivatives vanish when $e_i = B = 0$. Two v variations have weight 2η, and mixed variations have weight at least η, since $\mu > \eta$. Scaled Hölder product estimates prove membership in Y and (98).

On the cutoff annuli and exterior, radii are bounded below by fixed r_0, so ordinary Hölder matrix estimates apply. Globally, (95) gives

$$\max_{j\le k} \sup \varrho_p^j |\nabla^j (\mathrm{i}\partial\bar\partial S_p(x))|_{\widetilde{g}_p^R} \le C\|x\|_E. \tag{100}$$

Shrinking the parameter ball gives positivity. These arguments give a C^2 map and the stated uniform Lipschitz derivative.

9.3. **The global volume constraint.** Choose a nonnegative exterior bump supported away from the χ_i and positive on an open set. Normalize it to $\int a_p \, \mathrm{d}V_{\widetilde{g}_p^R} = 1$. Its Y norm is uniformly bounded. The projection

$$\Pi_p f = f - a_p \int_{X_p} f \, \mathrm{d}V_{\widetilde{g}_p^R}$$

is bounded from Y_p^R to $Y_{p,0}^R$, because the total volume is bounded and the weight is positive. Let $\mathcal{N}_p = \Pi_p \mathcal{M}_p^R$. Its derivative at zero is

$$D\mathcal{N}_p(0) = P_p^R J_p : E_p^R \longrightarrow Y_{p,0}^R.$$

Theorem 8.1 gives a uniform inverse. The contraction

$$x \longmapsto (D\mathcal{N}_p(0))^{-1} \left[\Pi_p F_p^R - \left(\mathcal{N}_p(x) - D\mathcal{N}_p(0)x\right)\right] \tag{101}$$

therefore solves $\mathcal{N}_p(x) = \Pi_p F_p^R$ for sufficiently small $\|F_p^R\|_Y$, with a threshold independent of R, p. More explicitly, the ball of radius $2C_{\mathrm{inv}}\|\Pi_p F_p^R\|_Y$ is preserved, and its Lipschitz constant is at most one half once this radius is below the fixed threshold from (98). Hence

$$\|x_p^R\|_E \leq C\|F_p^R\|_Y. \tag{102}$$

The projected equation implies $\mathcal{M}_p^R(x_p^R) - F_p^R = ca_p$. Both unprojected volume integrals equal V: one by (80), the other because $S_p(x_p^R)$ is a global potential in the same Kähler class. The function

$$c \longmapsto \int_{X_p} e^{F_p^R + ca_p} (\widetilde{\omega}_p^R)^3$$

is strictly increasing, since a_p is nonnegative and positive on a set of positive volume. Thus $c = 0$. We have solved the full Monge–Ampère equation with its single global compatibility condition.

9.4. **Smallness of the actual residual and uniqueness.** Proposition 7.1 gives the stronger weighted residual estimate

$$\limsup_{p\to 0} \|F_p^R\|_{C_\eta^{k,\alpha}} \leq C_k R^{\mu-\eta}. \tag{103}$$

On the core, (77) gives $R^{3/2-\eta}$. On the radius-R transition its scaled derivatives are $O(R^\mu)$ and the weight contributes $R^{-\eta}$. On the fixed exterior the residual tends smoothly to zero. One further derivative gives the Hölder bound. Since $\mu \leq 3/2$, these estimates prove (103) on the full admissible domain.

Choose R small enough for the uniform contraction threshold, and then p small at that fixed R. Set $u_p^R = S_p(x_p^R)$. Equations (100), (102) and (103) prove (92). The resulting positive form has volume $\kappa(p)\Theta_p$, so is Ricci-flat. Its class is exactly $c_1(\mathcal{L}_p)$. Uniqueness in the fixed Kähler class identifies it with ω_p. Elliptic bootstrapping makes it smooth, and repeating the argument at any higher fixed order estimates the same metric by uniqueness.

Combining (92) with (81) proves (93). It covers every cap, intervening annulus and exterior point. The actual metric is independent of the auxiliary radius R. Taking successively smaller fixed R also proves smooth convergence to g_0 on every fixed exterior compact set and to the normalized unit Stenzel metric on every fixed rescaled cap compact set. This completes the proof of Theorem 9.1.

10. The actual Hodge spectrum and its exterior marking

We now prove the spectral and form assertions of Theorem 1.1. All norms in this section are the geometric, unnormalized L^2 norms. On each smooth fibre, the real Hodge Laplacian has its usual H^2 operator domain and H^1 quadratic-form domain. Put

$$L_{\min}(p) = \min_i |\log|\tau_i(p)||, \qquad \mu_j(p) = \mathrm{Eig}_j(E_p, M), \quad j = 1, 2.$$

10.1. **Multiplicative transfer and the complete count.** We recall the variational metric comparison in a form which avoids differentiating the metric. If two metrics g, h on a closed six-manifold satisfy $(1-\delta)g \le h \le (1+\delta)g$, their squared L^2 norms in every degree differ by factors $1+O(\delta)$. The differential d and its graph domain are unchanged. On $\operatorname{Dom} \mathrm{d}/\ker \mathrm{d}$, the quotient norms differ by the same factors, since each is an infimum over the same closed subspace. Min–max for

$$\frac{\|\mathrm{d}u\|^2}{\|[u]\|^2}$$

therefore compares every ordered coexact eigenvalue multiplicatively. Exact positive eigenvalues in degree two are coexact positive eigenvalues in degree one. Taking the ordered union gives the full positive degree-two comparison. This is the variational comparison recalled in [W26a, Section 6]; it needs no estimate of the difference of codifferentials from a C^0 metric bound.

Fix R, A. Theorem 9.1 compares the actual metric with the model by an error whose limiting bound tends to zero as $R \downarrow 0$, $A \to \infty$. The model theorem gives exactly two small positive eigenvalues, relatively asymptotic to $\mathrm{Eig}_j(E_\varepsilon, M_R)$. Its complementary one-form and two-form gaps may depend on R, A. The actual normalized scales give

$$\log \frac{R}{2A\varepsilon_i(p)} = \frac{1}{3}|\log|\tau_i(p)|| + O_{R,A}(1). \tag{104}$$

Since the link norm is $2\pi/3$, this implies relative equivalence of E_ε and E_p, with error $O_{R,A}(L_{\min}^{-1})$. This is a comparison of each nonnegative summand, so it is independent of the condition number. The quotient masses M_R tend relatively to M under their common identification with V/K. Generalized min–max now gives

$$\lambda^+_{2,j}(p) = (1+o(1))\mu_j(p), \qquad j = 1, 2, \tag{105}$$

first at fixed R, A and then in the ordered limits. For each error tolerance the restriction on p applies throughout the full smooth locus.

Fixing one sufficiently accurate pair R, A transfers the complementary gaps. A gap uniform in all auxiliary choices is unnecessary. Topology gives $b_2(X_p) = 2$ throughout the smooth family. Thus the full low two-form space consists of two harmonic forms and two positive modes, with no additional cap or exterior modes.

10.2. **Complex type and primitivity.** The parallel holomorphic volume form on a Ricci-flat Kähler threefold, together with the metric, gives a parallel SU(3)- equivariant isomorphism $\Lambda^{0,1} \to \Lambda^{2,0}$. It intertwines the Hodge Laplacians: the connection terms agree under the parallel map, and the curvature terms are the same holonomy representation action. A nonzero positive $(2,0)$ eigencomponent below the one-form gap would therefore give a forbidden $(0,1)$ eigenform. Reality excludes the conjugate component as well.

The Kähler contraction Λ commutes with the Laplacian. If Λu were nonzero for a small positive eigenform, its differential would be a nonzero one-form eigenvector at the same eigenvalue. Similarly δu must vanish. Hence every small positive form is primitive, of real type $(1,1)$, and $u = \lambda^{-1}\delta \mathrm{d}u$ is coexact. This applies to the entire space S_p.

10.3. **A fixed observer and capture of the whole cluster.** Choose the compact exterior C_* from Section 1, with its central metric. Let Π_* be its L^2 projection onto distributionally closed two-forms with unrestricted tangential trace. The first-order estimate and harmonic cohomology map in Section 4 make

$$\mathcal{A}u = \mathrm{pr}_W^G[\Pi_*(u|_{C_*})] \tag{106}$$

a bounded operation on exterior L^2. The exterior identifications are fixed. It annihilates every globally closed form on X_p, whose restriction class lies in K. On a central harmonic representative η_D, it gives $\mathrm{pr}_W^G D$. The cohomology coefficients can be read from periods of the closed projection; we do not integrate an arbitrary unclosed L^2 form over cycles.

Compare the model metric g with the actual metric h under the surgery maps. If their inner products differ relatively by δ, their projections onto the orthogonal complement of the same closed space $\ker \mathrm{d}$ differ in norm by $O(\delta)$. In particular, for $u \perp_h \ker \mathrm{d}$ and $v = P_g u$,

$$\|u - v\|_g \le C\delta \|u\|_g, \qquad \mathrm{d}v = \mathrm{d}u. \tag{107}$$

Indeed for $z \in \ker \mathrm{d}$, the inner product $\langle u, z\rangle_g$ is just the difference of the two metric inner products and is bounded by $C\delta\|u\|_g\|z\|_g$. Taking a supremum over unit z proves the estimate. The subtracted form is closed.

If $u \in S_p$ is a unit vector, its energy is at most $C/L_{\min}$. The model energy of v is exactly $\|\mathrm{d}v\|_g^2$. The complementary model gap bounds the component of v outside the model small coexact space by $C_{R,A} L_{\min}^{-1/2}$ in norm. Thus u is within $C\delta + o_{R,A}(1)$ of that entire model space. Section 5 identifies it with W_R through its trials and their mass convergence. On these trials the fixed observer tends to

$$J_R : W_R \longrightarrow W, \qquad D \longmapsto \mathrm{pr}_W^G D.$$

Their central limits are closed representatives in class D, so the assertion holds in operator norm on the two-dimensional input space. The transported mass tends to M as $R \downarrow 0$. Consequently

$$\limsup_{p \to 0} \sup_{\substack{u \in S_p \\ \|u\|_h = 1}} |M(\mathcal{A}u, \mathcal{A}u) - 1| \le C\delta(R, A) + e_M(R), \qquad e_M(R) \to 0. \tag{108}$$

The observer constants remain bounded for small R because C_* lies outside all modified central regions and the quotient masses tend to a positive form.

Taking the ordered limits proves uniform mass convergence and injectivity of $\mathcal{A}$ on S_p. Since both dimensions are two, it is an isomorphism. In the notation of (7), polarization gives

$$(1 - e(p))M \le N_p \le (1 + e(p))M, \qquad e(p) \to 0. \tag{109}$$

No internal gap or eigenvalue-ratio assumption has entered this argument.

10.4. **All surviving harmonic classes.** For $k \in K$, let $h_g(k)$ and $h_h(k)$ be model and actual harmonic representatives of the corresponding smoothing class. Their difference $e = h_h(k) - h_g(k)$ is exact. Using their respective orthogonality to exact forms gives

$$\begin{aligned} \|e\|_h^2 &= -\langle h_g(k), e\rangle_h \\ &= -\big(\langle h_g(k), e\rangle_h - \langle h_g(k), e\rangle_g\big). \end{aligned}$$

Hence $\|e\|_h \le C\delta\|h_g(k)\|_g$. The model harmonic mass converges to $G_R|_K$, which tends to $G|_K$. The ordered limits prove the claimed actual harmonic mass on all of K. In particular both surviving classes, not just the polarizing class, are retained.

10.5. **The weighted lower bound in the fixed marking.** The relative eigenvalues and observed mass do not alone identify the marked energy. We first prove a one-sided form inequality. For an actual coexact u take $v = P_g u$ as above. Since $u - v$ is globally closed,

$$\mathcal{A}u = \mathcal{A}v. \tag{110}$$

Let C_R be the model exterior, Π_R its closed projection, and $\mathcal{A}_R$ its quotient observer. For small R, $C_* \subset C_R$. Set $\zeta = \Pi_R(v|_{C_R})$. Its class restricts unchanged to C_*, so $\mathcal{A}\zeta = J_R\mathcal{A}_R v$. The first-order projection estimate gives

$$|\mathcal{A}v - J_R\mathcal{A}_R v| \le C_R \|\mathrm{d}v\|_g. \tag{111}$$

Indeed $v - \zeta$ has $H^1(C_R)$ norm bounded by that right side, and the fixed observer is bounded on $L^2(C_*)$. This derivative estimate is stronger than mere convergence of the two observers.

The largest eigenvalue of E_p in any fixed Euclidean coordinates is $O(L_{\min}^{-1})$. Also $E_p(J_R D) = E_p(D)$, since the restriction periods vanish on K. Combine (56), (104) and (111) in the energy seminorm to obtain

$$\begin{aligned} E_p(\mathcal{A}u)^{1/2} &\le \big(1 + o_{R,A}(1) + C_R L_{\min}^{-1/2}\big)\|\mathrm{d}v\|_g \\ &\le \big(1 + C\delta(R,A) + o_{R,A}(1)\big)\|\mathrm{d}u\|_h. \end{aligned}$$

Every error is proportional to the input's own energy norm. Taking ordered limits, restricting to S_p and transporting by $\mathcal{A}$ gives

$$E_p \le (1 + e_1(p))Q_p, \qquad e_1(p) \to 0. \tag{112}$$

10.6. **Relative control of the full energy form.** Under the observation isomorphism, the generalized eigenvalues of (Q_p, N_p) are the two actual eigenvalues. Therefore (105) and (109) imply

$$\frac{\det Q_p}{\det E_p} = \frac{\lambda_{2,1}^+ \lambda_{2,2}^+}{\mu_1 \mu_2} \frac{\det N_p}{\det M} \longrightarrow 1. \tag{113}$$

Each eigenvalue of $B_p = E_p^{-1/2} Q_p E_p^{-1/2}$ is at least $(1+e_1(p))^{-1}$ by (112), while their product tends to one by (113). The larger is at most $\det(B_p)(1 + e_1(p))$. Thus both tend to one, which proves

$$(1 - e(p))E_p \le Q_p \le (1 + e(p))E_p. \tag{114}$$

This determinant argument never bounds E_p^{-1} in a fixed norm.

For an actual orthonormal eigenbasis, let C_p have columns $\mathcal{A}u_j$. Then $C_p^{\mathsf{T}} N_p C_p = I$ and $C_p^{\mathsf{T}} Q_p C_p = \Lambda_p$. Conjugating (109) and (114) gives the two matrix estimates in (9). In particular the (i,j) energy error is bounded by $e(p)\sqrt{\lambda_{2,i}^+ \lambda_{2,j}^+}$. This remains meaningful for arbitrarily separated eigenvalues.

Compactness gives a uniform error tending to zero, without an explicit rate. The eigenline and separated-path assertions are proved next.

11. Successive scales and relative spectral transport

The relative form estimates determine more than the ordered small eigenvalues. They transport groups of marked eigenspaces whenever those groups have a relative gap. For fixed restriction maps, the limiting groups have a direct description by successive kernels.

11.1. **Successive kernels of fixed maps.** Let V be a finite-dimensional real space with positive mass M. For fixed linear maps d_j to Euclidean spaces, put $B_j(x) = \|d_j x\|^2$ and

$$K_0 = V, \quad K_j = \bigcap_{i \le j} \ker d_i, \quad V_j = K_{j-1} \cap K_j^{\perp_M}, \quad r_j = \dim K_{j-1} - \dim K_j.$$

Zero values of r_j are allowed. Suppose $t_j > 0$, $t_{j+1}/t_j \to 0$, and $E_p = \sum_{j=1}^{s} t_j B_j$. Let Q_p, N_p satisfy

$$(1-e_p)E_p \le Q_p \le (1+e_p)E_p, \qquad (1-e_p)M \le N_p \le (1+e_p)M, \qquad e_p \to 0. \tag{115}$$

Theorem 11.1. *For each j with $r_j > 0$, precisely r_j positive eigenvalues of (Q_p, N_p) have scale t_j. Dividing them by t_j gives, with multiplicity, the positive generalized eigenvalues of*

$$(B_j|_{K_{j-1}}, M|_{K_{j-1}}). \tag{116}$$

The exact kernel has dimension $\dim K_s$. The cumulative spectral space of all groups with index greater than j, together with the kernel, tends to K_j. The j-th group tends to V_j in the M-Grassmannian.

Proof. For fixed j set $S_{p,j} = (N_p + Q_p/t_j)^{-1} N_p$. The vector $u = S_{p,j} f$ solves

$$N_p(u,x) + t_j^{-1} Q_p(u,x) = N_p(f,x). \tag{117}$$

Testing with $x = u$ bounds $\|u\|_M$ and $E_p(u)/t_j$ by constants times $\|f\|_M$ and $\|f\|_M^2$, respectively. For $j > 1$, positivity of $\sum_{i<j} B_i$ on $K_{j-1}^{\perp_M}$ then gives

$$\operatorname{dist}_M(u, K_{j-1})^2 \le C(t_j/t_{j-1})\|f\|_M^2.$$

For $j = 1$ there is no excluded complement.

The form estimate implies its bilinear version

$$|(Q_p - E_p)(u,x)| \le e_p \sqrt{E_p(u)}\sqrt{E_p(x)}. \tag{118}$$

This follows by passing to the quotient by $\ker E_p$ and conjugating by its positive square root. For $x \in K_{j-1}$, all $B_i(u,x)$ with $i < j$ vanish exactly, and $E_p(x)/t_j$ is bounded. Divide (117) by the scales as written and pass to a subsequence. Every limit $u \in K_{j-1}$ satisfies

$$M(u,x) + B_j(u,x) = M(f,x), \qquad x \in K_{j-1}.$$

This equation has a unique solution. Finite dimension therefore gives operator norm convergence of $S_{p,j}$ to the M-orthogonal compression of $(I + B_j|_{K_{j-1}})^{-1}$, extended by zero on the complement. Conjugating by the positive mass square roots turns this into ordinary self-adjoint norm convergence.

The eigenvalues of $S_{p,j}$ are $(1 + \lambda/t_j)^{-1}$. Their limits are zero on the fast complement, the r_j values strictly between zero and one specified by (116), and one on K_j. This proves the scale and coefficient assertions. Repeating for all j accounts for every dimension. The exact kernel is K_s by (115).

The isolated limit eigenvalue one gives convergence of the cumulative slow projector to the projector onto K_j. Differences of consecutive cumulative projectors give the projector onto $K_{j-1} \cap K_j^{\perp_M}$. This proves the subspace assertions, including steps with $r_j = 0$. □

The pair (116) can equivalently be taken on K_{j-1}/K_j, with its quotient mass $\inf_{k \in K_j} M(x+k)$. If positive row weights satisfy $w_e/t_j \to c_e \in (0,\infty)$ within each group, use $B_j = \sum_{e \text{ in group } j} c_e a_e^* a_e$. The relative error of this positive sum tends to zero without dividing by its smallest weight. Thus logarithmic and iterated-logarithmic scales are retained. The

grouping is attached to a sector or path with those limits. It need not define one filtration on the entire multiparameter base.

11.2. **A relative-gap estimate.** At a single parameter, let $E \ge 0$, $M > 0$ on a space of dimension at most n, and suppose Q, N obey

$$(1-e)E \le Q \le (1+e)E, \qquad (1-e)M \le N \le (1+e)M, \qquad 0 < e < 1/4.$$

Order their generalized eigenvalues γ_i and λ_i in the same direction, including zeros. A consecutive positive index set I is relatively separated by κ if $0 < \kappa \le 1$ and

$$|\gamma_i - \gamma_j| \ge \kappa \max(\gamma_i, \gamma_j), \qquad j \in I,\ i \notin I. \tag{119}$$

Let F_I and G_I be the corresponding finite and observed subspaces.

Lemma 11.2. *If $e \le c_n \kappa$, then*

$$\|P^M_{F_I} - P^M_{G_I}\|_M \le C_n e/\kappa. \tag{120}$$

The exact kernels coincide. There is no factor depending on the positive spectral condition number.

Proof. Choose an M-orthonormal eigenbasis u_i of (E, M) and an N-orthonormal eigenbasis v_j of (Q, N). Min–max gives λ_j/γ_j between $(1-e)/(1+e)$ and $(1+e)/(1-e)$ for positive eigenvalues, with zeros preserved. Testing the v_j equation against u_i yields

$$(\gamma_i - \lambda_j) M(u_i, v_j) = -(Q-E)(u_i, v_j) + \lambda_j (N-M)(u_i, v_j).$$

Since $E(v_j) \le \lambda_j/(1-e)$ and $M(v_j) \le (1-e)^{-1}$, the right side is at most $Ce(\sqrt{\gamma_i \lambda_j} + \lambda_j)$ in absolute value. For $j \in I$, $i \notin I$, (119) and smallness of e/κ give

$$|\gamma_i - \lambda_j| \ge c\kappa \max(\gamma_i, \lambda_j).$$

Every off-block coefficient is therefore at most Ce/κ. A unit combination of at most n columns has off-block norm at most $C_n e/\kappa$, after using mass equivalence. Equal dimensions and the principal-angle identity give (120). Explicitly, an M-orthonormal basis of G_I whose off-block norm is at most a has projection to F_I with least singular value at least $\sqrt{1-a^2}$; its largest principal-angle sine is at most a. Kernel equality follows directly from the two-sided energy inequality. □

Through (106), the lemma compares marked actual subspaces with the independently constructed finite subspaces. It does not identify forms on unrelated manifolds without that map. Individual vectors require an internal relative gap; repeated leading eigenvalues give subspace assertions.

11.3. **Limits of the fixed-map description.** The fixed-map hypothesis has mathematical content. With identity mass and target weight, the moving map

$$d_h = \begin{pmatrix} 1 & 0 \\ 1 & h^m \end{pmatrix}, \qquad m > 0,$$

has row norms bounded away from zero, but $d_h^* d_h$ has trace $2 + h^{2m}$ and determinant h^{2m}. Its eigenvalues are asymptotic to 2 and $h^{2m}/2$. Multiplying the energy by $t(h) \to 0$ gives two small eigenvalue scales $2t$ and $th^{2m}/2$. The rate at which the rows become dependent is additional data. Replacing the lower-right entry by $h^m - h^m + h^n$, $n > m$, replaces the second exponent by $2n$; cancellation must be performed before assigning orders to minors.

The mass is equally necessary. For $E_h = h^2 I$ and $M_h = \mathrm{diag}(h^2, 1)$, the generalized eigenvalues are $1, h^2$. Vanishing energy on both coordinate vectors does not make both eigenvalues

tend to zero. Finally, orthogonal conjugates of $\operatorname{diag}(h, h^2)$ have identical characteristic coefficients, but a nonconvergent rotation angle produces nonconvergent lines. Eigenvalues alone do not determine marked spectral subspaces.

For the conifold theorem the restriction rows are fixed and the central mass is positive definite. Thus the hypotheses of Theorem 11.1 are verified on every capacity-ratio sector. The two-dimensional pair also has a relative gap throughout the full domain, which gives a stronger pointwise eigenline statement.

12. Conifold eigenlines and changes of choices

12.1. **Two simple lines on the full domain.** Put $c = w_f/w_g > 0$. The finite characteristic polynomial is

$$256\mu^2 - 36(w_f + w_g)\mu + 5w_f w_g = 0. \tag{121}$$

Its roots are $\mu_\pm = w_g \kappa_\pm(c)$, where

$$\kappa_\pm(c) = \frac{9(c+1) \pm \sqrt{81(c-1)^2 + 4c}}{128}. \tag{122}$$

Since $81(c-1)^2 + 4c - (c+1)^2 = 80(c-1)^2$,

$$0 < \mu_-/\mu_+ \le \frac{4}{5}, \qquad \mu_+ - \mu_- \ge \frac{1}{5}\mu_+. \tag{123}$$

Thus the finite lines are relatively separated throughout the positive quadrant. The actual relative eigenvalue estimate makes both actual positive eigenvalues simple for all sufficiently small admissible p, uniformly in c. Lemma 11.2 then gives

$$\|P^M_{\mathcal{A}\ell_\pm(p)} - P^M_{L_\pm(p)}\|_M \le Ce(p) \to 0, \tag{124}$$

where $\ell_\pm$ are the actual real eigenlines and $L_\pm$ the generalized finite eigenlines. No sign for an eigenvector is chosen.

An exact expression is

$$L_\pm = \mathbb{R}(4\mu_\pm,\ 36\mu_\pm - 5w_f). \tag{125}$$

The vector is nonzero and satisfies $(E_p - \mu_\pm M)x = 0$. For $c \to \infty$,

$$\begin{aligned} &\mu_+ \sim \frac{9}{64} w_f, && \mu_- \sim \frac{5}{36} w_g, \\ &L_+ \longrightarrow \mathbb{R}(1, 1/9), && L_- \longrightarrow \mathbb{R}(0, 1). \end{aligned} \tag{126}$$

For $c \to 0$ the roles reverse:

$$\mu_+ \sim \frac{9}{64} w_g, \quad \mu_- \sim \frac{5}{36} w_f, \qquad L_+ \to \mathbb{R}(1/9, 1), \quad L_- \to \mathbb{R}(1, 0).$$

For $c \to c_0 \in (0, \infty)$ both have scale w_g, with constants $\kappa_\pm(c_0)$ and the corresponding distinct finite lines. At $c = 1$ the lines are $\mathbb{R}(1, -1)$ and $\mathbb{R}(1, 1)$, with coefficients $1/8$ and $5/32$, respectively.

The boundary filtration for $c \to \infty$ has slow subspace $\mathbb{R}g$ and its M-orthogonal fast complement $\mathbb{R}(1, 1/9)$. These are exactly the successive-kernel spaces in Theorem 11.1. A path with no ratio limit need not have a limiting filtration; the pointwise statement (124) still applies.

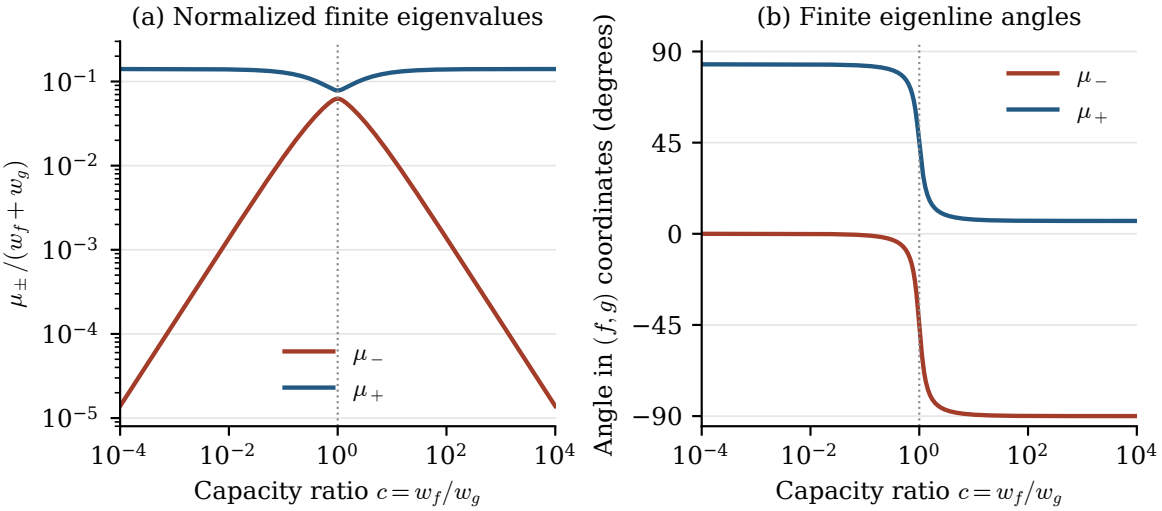


FIGURE 1. The finite pair $(\mathrm{diag}(c,1),M)$, with quotient mass M. Left: its generalized eigenvalues divided by $1+c$. Right: the angles of eigenline representatives with positive first coordinate in the displayed (f,g) basis. The dotted line marks equal capacities.

12.2. **Realized logarithmic hierarchies.** For positive functions $F(h), G(h) \to \infty$, the admissible slice

$$p(h) = (a,a,a,-3a;\ b,b,b,-3b), \qquad a = e^{-F(h)}, \quad b = e^{-G(h)} \tag{127}$$

eventually belongs to B°. The bounded holomorphic units and the factor -3 change each logarithm by a bounded amount. Hence

$$w_f \sim \frac{8\pi}{F(h)}, \qquad w_g \sim \frac{8\pi}{G(h)}.$$

If $F/G \to 0$, the actual simple eigenvalues satisfy

$$\lambda_+ \sim \frac{9\pi}{8F(h)}, \qquad \lambda_- \sim \frac{10\pi}{9G(h)}. \tag{128}$$

Taking $F = h^{-1}$, $G = h^{-2}$ gives the constants in (12). Taking $G = h^{-1}\log(1/h)$ instead gives the orders h and $h/\log(1/h)$; taking $G = h^{-1}\log\log(1/h)$ gives h and $h/\log\log(1/h)$. These are paths on the actual family; they are not holomorphic arcs at the origin. This completes the eigenline and path consequences announced after Theorem 1.1.

12.3. **Fixed cutoffs and exterior identifications.** Keep W, M fixed, but choose two compact central exteriors C_1, C_2 with their central-metric closed projections and cohomology maps. Let $\mathcal{A}_1, \mathcal{A}_2$ be the resulting observers. Choose a larger compact exterior C containing both, and set $\zeta = \Pi_C u$. The class of this closed form agrees on both smaller exteriors, so $\mathcal{A}_1\zeta = \mathcal{A}_2\zeta$. The first-order projection estimate gives

$$\|(\mathcal{A}_2 - \mathcal{A}_1)u\|_M \le C\|\mathrm{d}u\|_{L^2(C,g_p)}. \tag{129}$$

Here the actual and central exterior metrics are uniformly equivalent. The constants may depend on these fixed choices.

On S_p, put $B_p = \mathcal{A}_2(\mathcal{A}_1|_{S_p})^{-1}$ and $\sigma_p = \max \mathrm{Eig}(E_p, M)$. Equation (129) and the mass and energy estimates show that $\mathcal{A}_2$ is also invertible for small p, and

$$\begin{aligned} \|B_p - I\|_M &\le C\sqrt{\sigma_p}, \\ \sqrt{E_p((B_p - I)x)} &\le C\sqrt{\sigma_p}\sqrt{E_p(x)}. \end{aligned} \tag{130}$$

For the second line, use $E_p(y) \le \sigma_p M(y)$ and $q_p(\mathcal{A}_1^{-1}x) \le (1 + o(1))E_p(x)$. The first follows from the cluster bound $q_p \le C\sigma_p \|\cdot\|^2$. The triangle inequality in the energy seminorm gives

$$(1 - C\sqrt{\sigma_p})E_p \le B_p^* E_p B_p \le (1 + C\sqrt{\sigma_p})E_p,$$

after changing C, and the mass obeys the analogous estimate. The observed forms transform exactly as $B_p^* Q_{2,p} B_p = Q_{1,p}$ and $B_p^* N_{2,p} B_p = N_{1,p}$. Thus both observers give the same relative description without a lower bound on the eigenvalue ratio.

The same conclusion holds when exterior identifications differ by a smooth isotopy preserving the relevant exterior, with uniform derivative, inverse-Jacobian and generating-vector-field bounds. The homotopy formula reads $F_1^* u - F_0^* u = \mathrm{d}Ku + K\mathrm{d}u$. The observation of the exact term is zero, and the other term is bounded by $C\|\mathrm{d}u\|$. Density extends the argument to form-domain inputs. An arbitrary shrinking cutoff or unbounded isotopy requires another estimate. Phase interpolations supported in the necks leave the fixed exterior observer unchanged.

12.4. **Coordinates, polarization and base change.** Under a finite coordinate change $x = Sx'$, the data transform as

$$E' = S^* E S, \qquad M' = S^* M S, \qquad \mathcal{A}' = S^{-1}\mathcal{A}.$$

Generalized eigenvalues and intrinsic relative inequalities are unchanged, and subspaces are transported by S. No Euclidean condition-number bound on S is needed. An integral marking change must preserve the specified lattice; this observation does not supply a lattice from spectral data.

A permitted analytic node-coordinate change has the form $\tau_i' = u_i(p)\tau_{\pi(i)}$, with holomorphic units bounded above and away from zero. Transport the link generators by their signs and permutation. Then

$$|\log|\tau_i'|| = |\log|\tau_{\pi(i)}|| + O(1),$$

so E_p' and E_p differ relatively by $O(L_{\min}^{-1})$. This follows term by term from the reciprocal lengths and the nonnegative period squares. The geometric mass transforms by congruence. Lemma 11.2 and (123) give an $O(L_{\min}^{-1})$ change of the transported finite eigenlines.

Polarization is genuine geometric input. Scaling the Kähler class by a fixed $c > 0$ scales the Ricci-flat metric by c. In real dimension six the two-form squared L^2 norm scales by c, whereas its d-energy scales by one. The eigenvalues therefore scale by c^{-1}.

For a holomorphic base map with $\tau_i(s) = a_i s^{\nu_i} + O(s^{\nu_i+1})$, $a_i \neq 0$ and $\nu_i > 0$,

$$\frac{2\pi}{|\log|\tau_i(s)||} \sim \frac{2\pi}{\nu_i |\log|s||}. \tag{131}$$

All capacities have one logarithmic order. A ramification $s = v^m$ multiplies every ν_i by m and divides the leading coefficients expressed in $|\log|v||$ by m. This reparametrizes fibres; it does not preserve a coefficient in a fixed parameter. For an admissible monomial map $\tau_i(z) = u_i(z)\prod_j z_j^{a_{ij}}$, the corresponding length is $\sum_j a_{ij}|\log|z_j|| + O(1)$. Such a map must factor through the actual relation base and smooth the selected nodes. Eight arbitrarily prescribed monomials need not do so.

13. Attachment maps, monodromy and the spectral hierarchy

13.1. **Relations among vanishing cycles.** Let Y be the small resolution, U the common exterior, C_i the exceptional curves, and δ_i the oriented vanishing spheres. The integral Thom and Mayer–Vietoris calculation in [W26a, Section 3.3, proposition on restriction and spectral coefficient] gives

$$\begin{aligned} 0 \longrightarrow H^2(X_p;\mathbb{Z}) \longrightarrow H^2(Y;\mathbb{Z}) \xrightarrow{r} \mathbb{Z}^8 \xrightarrow{\partial} H^3(X_p;\mathbb{Z}) \\ \longrightarrow H^3(U;\mathbb{Z}) \longrightarrow 0, \end{aligned} \tag{132}$$

where $r(D)_i = D \cdot C_i$ and $\partial e_i = \mathrm{PD}_L(\delta_i)$. The latter class is defined by the oriented Thom construction, including any torsion in its global image.

Over $\mathbb{Q}$, put $Q_{\mathrm{lost}} = H^2(Y;\mathbb{Q})/\operatorname{im} H^2(X_p;\mathbb{Q})$ and $V_{\mathrm{van}} = \operatorname{im}\partial$. Then

$$0 \longrightarrow Q_{\mathrm{lost}} \xrightarrow{r} \mathbb{Q}^8 \xrightarrow{\partial} V_{\mathrm{van}} \longrightarrow 0. \tag{133}$$

Their dimensions are two and six, respectively. In our quotient basis the columns of r are $(1,1,1,1;0,0,0,0)$ and $(0,0,0,0;1,1,1,1)$. Thus the lost degree-two cohomology quotient is the space of relations among the cohomological vanishing cycles, and its composite to V_{van} is zero. The exceptional-curve map $s : \mathbb{Q}^8 \to H_2(Y;\mathbb{Q})$, $s(e_i) = [C_i]$, is dual to r. Its six-dimensional relation space $\ker s$ is a different space, paired with the period coordinates of Section 2.

Lee–Lin–Wang realize (133) as the rational weight-two basic exact sequence of a semistable model [LLW18, Theorem 1.14]; their Proposition 3.16 treats vanishing-period coordinates. These are classical deformation and Hodge-theoretic inputs. Rational exactness does not by itself establish integral saturation or recover an unknown attaching map.

13.2. **The nearby-cycle sequence.** For a specified proper one-parameter ordinary-double-point smoothing f with smooth total space, the unshifted nearby- and vanishing-cycle triangle is

$$i^*\mathbb{Q} \longrightarrow R\psi_f\mathbb{Q} \longrightarrow R\phi_f\mathbb{Q} \longrightarrow i^*\mathbb{Q}[1].$$

The Milnor fibres of the threefold nodes give the specialization segment

$$\begin{aligned} 0 \longrightarrow H^3(X_0;\mathbb{Q}) \longrightarrow H^3(X_t;\mathbb{Q}) \xrightarrow{\mathrm{can}} \mathbb{Q}^8 \\ \longrightarrow H^4(X_0;\mathbb{Q}) \longrightarrow H^4(X_t;\mathbb{Q}) \longrightarrow 0. \end{aligned} \tag{134}$$

The map can evaluates on vanishing spheres. Its image annihilates the relations in (133). The map ∂ there is its Poincaré–excision dual, with the chosen orientation convention; it is not the arrow from $\mathbb{Q}^8$ to $H^4(X_0;\mathbb{Q})$ in (134). For the triangles, stalks and canonical/variation identities see [Mas14, Section 3], and for the conifold specialization see [LLW18, Section 1]. After ramified base change the total space need not be smooth; one must use nearby-cycle base change or a semistable model instead of applying the smooth-total-space local argument without its hypothesis.

13.3. **What monodromy does not determine.** Picard–Lefschetz gives $N_i\gamma = \pm\langle\gamma,\delta_i\rangle\delta_i$. The vanishing spheres are disjoint, so $N_iN_j = 0$. For an all-node loop with positive winding numbers ν_i,

$$T = I + N, \qquad N = \sum_i \nu_i N_i, \qquad N^2 = 0 \tag{135}$$

as in [LLW18, Section 3.3, equation (3.4)]. The monodromy image is abelian; no assertion that the fundamental group of the discriminant complement is abelian is needed.

The weight filtration centred at three is

$$0 = W_1 \subset W_2 = \operatorname{im} N \subset W_3 = \ker N \subset W_4 = H^3. \tag{136}$$

Here $\dim W_2 = 6$. These rational subspaces are independent of the positive ν_i. In a symplectic basis let B have columns δ_i and let J be the intersection matrix. Then $N = \pm BDB^{\mathsf{T}}J$, $D = \operatorname{diag}(\nu_i) > 0$. The matrix BDB^{T} has the same rank and image as B, so $\operatorname{im} N = \operatorname{im} B$ and $\ker N = (\operatorname{im} B)^{\perp_J}$. Integral image indices can change under ramification. The characterization of the weight filtration is [Sch73, Section 6, Lemma (6.4)]; the identification of V_{van} with W_2 is [LLW18, Lemma 1.12].

The two admissible paths

$$(a,b) = (e^{-h^{-1}}, e^{-h^{-2}}) \quad \text{and} \quad (a,b) = (e^{-h^{-1}}, e^{-h^{-3}})$$

have the same marked central attachment sequence and local monodromy representation. Choose the same reference all-node loop; its operator and weight filtration are then the same. Nevertheless (128) gives the actual spectral orders (h, h^2) and (h, h^3). The stated topology and monodromy data therefore do not determine the capacity hierarchy.

An interval path has no distinguished loop, and a nonanalytic path does not define an ordinary holomorphic-disc nearby-cycle functor. We make no assertion that the full limiting mixed Hodge structures or period asymptotics agree on these paths. The limiting Hodge filtration contains further information; its coordinate dependence by $\exp(cN)$ is recorded before [Sch73, Theorem (6.16)].

13.4. **Comparison with adiabatic spectral sequences.** For a fixed compact fibration, Mazzeo–Melrose identify the adiabatic Hodge problem with the Leray spectral sequence [MM90, Theorems (7), (15) and Corollary (18)]. For a fixed smooth Riemannian foliation on a closed manifold, Álvarez López–Kordyukov study $g_h = h^{-2} g_\perp + g_F$. The transverse-degree filtration gives

$$\mathrm{d}_h = \mathrm{d}_{0,1} + h\mathrm{d}_{1,0} + h^2 \mathrm{d}_{2,-1}.$$

Their Theorem A counts eigenvalues, including zeros, of order $O(h^{2k})$ by the degreewise dimension of the k-th page for $k \geq 2$; at $k = 1$ it uses the Hausdorff quotient of the first page [AK00]. The corrected Theorem B gives only weak subsequential convergence at $k = 1$ in its stated setting, with strong convergence for $k \geq 2$ under its energy hypotheses.

Those results include a filtered differential complex and its page differentials. The successive kernels in Theorem 11.1 come instead from the fixed restriction rows of a finite energy form. We have not constructed page differentials or a compatible de Rham Laplacian in every degree. The conifold smoothing also changes topology, unlike the fixed-foliation adiabatic metric. Thus neither the capacity groups nor the relation space in (133) are identified with those pages.

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